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About this copy

This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. N] shows where page N of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM15.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Misprint.** Proof of (1.1), page 116: θ_{k+1} and λ_{k+1} become $\theta_{\kappa+1}$ and $\lambda_{\kappa+1}$.
2. **Misprint.** Proof of (1.1), page 116: λ_{k+1} becomes $\lambda_{\kappa+1}$.
3. **Proof gap.** Proof of (2.1), page 116: the case $\bar{\Omega} = \emptyset$ is set aside first.
4. **Misprint.** Claim (8) in the proof of (2.1), page 118: $N(P_i \cup B)$ becomes $N(P_i \cap B)$.
5. **Misprint.** Proof of (2.1), page 119: $V(A \cup B)$ becomes $V(A \cap B)$.
6. **Statement mistake.** Statement of (3.4), page 122: the paths P_i are mutually disjoint and meet H only in their ends.
7. **Missing proof.** Proof of (3.4), page 123: the proof that (3.2)(iv) holds is added.

8. **Missing proof.** Proof of (3.4), page 123: the application of Graph Minors XII, (6.1), is given in full.
9. **Misprint.** Proof of (3.4), page 123: the indices of e_1, e_2 and of v_1, v_2 in the last sentence are corrected.
10. **Statement mistake.** Statement of (3.5), page 123: a hypothesis on the H-normal lines from $bd(\Lambda_1)$ to $bd(\Lambda_2)$ is added.
11. **Misprint.** Section 4, page 124: "radial graph of G" becomes "radial drawing of H".
12. **Degenerate case.** Statement of (5.2), page 129: $\mu \geq 0$ becomes $\mu \geq 1$.
13. **Misprint.** Claim (7) in the proof of (5.2), page 131: Λ_1 becomes Z_1.
14. **Misprint.** Claim (8) in the proof of (5.2), page 131: Λ_2 becomes Z_2.
15. **Misprint.** Statement of (6.2), page 134: $2\tau_i$ becomes $2\tau_1$.
16. **Proof gap.** Section 7, page 135: in the definition of π, $4 + \mu + \theta_{k+1}/2$ becomes $7 + \mu + \theta_{k+1}/2$.
17. **Proof gap.** Section 7, page 138: $\tau_2 = 3 + \theta'/2$ becomes $\tau_2 = 6 + \theta'/2$.
18. **Misprint.** Claim (8) in the proof of (7.7), page 142: $j \leq i$ becomes $j \leq k$.
19. **Proof gap.** Claim (1) in the proof of (7.11), page 143: $2\tau - 6$ becomes $2\tau - 12$.

[p. 112] # Graph Minors
XV. Giant Steps

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Let G be a graph with a subgraph H drawn with high representativity on a surface Σ. When can the drawing of H be extended "up to 3-separations" to a drawing of G in Σ if we permit a bounded number (κ say) of "vortices" in the drawing of G, that is, local areas of non-planarity? (The case $\kappa = 0$ was studied in the previous paper of this series.) For instance, if there is a path in G with ends in H, far apart, and otherwise disjoint from H, then no such extension exists. We are concerned with the converse; if no extension exists, what can we infer about G? It turns out that either there is a path as above, or one of two other obstructions is present.

1. Introduction

The current objective of this sequence of papers is to prove a result about the structure of the graphs not containing a fixed minor (K_n, say). The proof will be completed in the next paper, but here we accomplish a substantial part. Before we go into the details of what we are going to prove in this paper, it may be helpful if we sketch the bigger picture. Suppose that G has no K_n minor. If G has small tree-width, we are done; so let us assume it has large tree-width (at least some enormous function of n). Hence it has a planar subgraph H_0 with large tree-width, by a theorem of an earlier paper. Take a drawing of H_0 in a sphere Σ_0. The pair H_0, Σ_0 can be regarded as a degenerate case of the following: a subgraph H of G, drawn in a surface Σ, with large representativity (that is, every simple [p. 113] closed curve in the surface that does not bound a disc meets the drawing many times). Our initial pair H_0, Σ_0 is a degenerate case, because for the sphere

*This work was partially performed under a consulting agreement with Bellcore.

Σ_0 representativity does not make sense, but we showed in earlier papers that in the sphere, high tree-width is the appropriate analogue for high representativity. Let us make a sequence H_i, Σ_i $(i = 0, 1, 2, ...)$ of such pairs, at each stage sacrificing a large amount of the remaining representativity if necessary for an increase in genus (we permit Σ_i to be non-orientable, in which case its "genus" is half the number of crosscaps). Since H_0, Σ_0 has large representativity, the initial few terms in our sequence still have high representativity. Now every graph with high representativity that can be drawn in a surface in which K_n can be drawn has a K_n minor. Hence the process stops in the initial few terms, before Σ_i has grown complex enough that K_n can be drawn in it. We therefore have a pair H, Σ (say), where Σ has bounded genus and H is drawn in it with very large representativity, so that no improvement in Σ can be made.

How does the remainder of G attach to the subgraph H? The theorem that we are going to prove here says roughly that either

(i) we can delete a small area of H, and replace it by some other subgraph of G, to obtain a pair H', Σ' where Σ' is obtained by adding a crosscap to Σ and H' still has large representativity–but this contradicts the choice of H, Σ;

(ii) we can choose another H', Σ so that H' can be drawn in Σ with crossings, but with a large constant number of crossings, all pairwise "far apart" and still with high representativity in the appropriate sense–but this implies that G has a K_n minor, a contradiction;

(iii) we can rearrange a small area of H so that some path of G has both ends far apart in H and is otherwise disjoint from H; or,

(iv) G has the desired structure (roughly, all of G can be drawn in Σ "up to 3-separations," except for a bounded number of local areas of non-planarity, called "vortices").

Thus, we are done in every case except (iii). That case needs work, and to handle it we actually need a more complicated optimization of H, Σ than is described here; but we hope this provides some motivation for the present result.

This paper is closely related to [5], and uses most of the same terminology. In particular, the reader should see [5, Sections 2–4] for the meaning of the following terms and notation: $G_1 \cup G_2$, $G_1 \cap G_2$, separation, order of a separation, tangle, order of a tangle, free, open and closed disc, surface, $bd(\Sigma)$, O-arc, line, ends of a line, drawing, $U(H)$, region, $R(H)$, atom, $A(H)$, radical drawing, respectful tangle, metric of a tangle, λ-zone, clearing, 2-cell, rigid, dial, natural order, society, $\bar{\Omega}$, segregation, 3-segregation, [p. 114] $V(\mathscr{S})$, arrangement, $\mathscr{T}$-local, bridge, H-path in G (called an "H-

pathing" in [5], due to a typographical error).

Let (G, Ω) be a society, and let the elements of $\bar{\Omega}$ be $v_1, ..., v_n$, numbered in order under Ω. A transaction in (G, Ω) is a set $\mathscr{P}$ of mutually disjoint paths of G, each with distinct ends both in $\bar{\Omega}$, such that for some i, j with $0 \le i \le j \le n$, each member of $\mathscr{P}$ has exactly one end in $\{v_{i+1}, ..., v_j\}$. For $\rho \ge 0$, a ρ-vortex is a society with no transaction of cardinality $> \rho$. A segregation $\mathscr{S}$ of a graph G is of type (ρ, κ), where $\rho, \kappa \ge 0$ are integers, if there are at most κ members $(A, \Omega) \in \mathscr{S}$ with $|\bar{\Omega}| > 3$, and each is a ρ-vortex. Thus, segregations of type $(\rho, 0)$ are 3-segregations, for any ρ.

If $\mathscr{T}^*$ is a tangle in G, a segregation $\mathscr{S}$ of G is $\mathscr{T}^*$-central if for each $(A, \Omega) \in \mathscr{S}$ there is no $(A^*, B^*) \in \mathscr{T}^*$ with $B^* \subseteq A$. We denote the order of a tangle $\mathscr{T}$ by $ord(\mathscr{T})$. Let $\mathscr{T}^*$ be a tangle in G, and let Σ be a surface with $bd(\Sigma) = \emptyset$. A triple $H, \eta, \mathscr{T}$ is a Σ-span of order θ (in G, with respect to $\mathscr{T}^*$) if

(i) H is a rigid drawing in Σ

(ii) η is an isomorphism from H to a subgraph $\eta(H)$ of G

(iii) $\mathscr{T}$ is a respectful tangle in H of order θ

(iv) $\theta \le ord(\mathscr{T}^*)$, and $(A \cap \eta(H), B \cap \eta(H)) \in \eta(\mathscr{T})$, for every $(A, B) \in \mathscr{T}^*$ of order $< \theta$.

($\eta(\mathscr{T})$ denotes the image of $\mathscr{T}$ under η—we shall frequently use similar notation without further explanation.) Whenever possible we shall arrange that η is the identity function on H, and so H itself is a subgraph of G; and in that case we loosely say that $H, \mathscr{T}$ is a Σ-span (without reference to η or θ).

If H is a 2-cell drawing in Σ, and $\mathscr{T}$ is a respectful tangle in H with metric d, then for $\sigma \in \Sigma$ and $b \in A(H)$, we define $d(\sigma, b) = d(a, b)$ where $\sigma \in a \in A(H)$; and for $\sigma_1, \sigma_2 \in \Sigma$ we define $d(\sigma_1, \sigma_2) = d(a_1, a_2)$ where $\sigma_1 \in a_1 \in A(H)$ and $\sigma_2 \in a_2 \in A(H)$. If H' is another 2-cell drawing in Σ with a respectful tangle $\mathscr{T}'$ with metric d', we say that $\mathscr{T}'$ is a λ-compression of $\mathscr{T}$ (where $\lambda \ge 0$ is an integer) if $d'(\sigma_1, \sigma_2) \ge d(\sigma_1, \sigma_2) - \lambda$ for all $\sigma_1, \sigma_2 \in \Sigma$, and $ord(\mathscr{T}') = ord(\mathscr{T}) - \lambda$.

If $\Sigma, H, \eta, \mathscr{T}, H', \eta', \mathscr{T}', G^*, \mathscr{T}^*$ are as above, and $z \in A(H)$, we say that $H', \eta', \mathscr{T}'$ is obtained from $H, \eta, \mathscr{T}$ by rearranging within λ of z, where $\lambda \ge 0$ is an integer, if

(i) $a \in A(H')$ for all $a \in A(H)$ with $d(z, a) > \lambda$, where d is the metric of $\mathscr{T}$,

(ii) $\eta(x) = \eta'(x)$ for every vertex or edge x of H with $d(z, x) > \lambda$, and

(iii) $\mathscr{T}'$ is a $(4\lambda + 2)$-compression of $\mathscr{T}$.

[p. 115] A Σ-span $H, \eta, \mathscr{T}$ is μ-stepped if there is an $\eta(H)$-path in G with ends $\eta(u), \eta(v)$ such that $d(u, v) \ge \mu$, where d is the metric of $\mathscr{T}$. A Σ-span

$H, \eta, \mathscr{T}$ is (λ, μ)-flat, where $\lambda, \mu \geq 0$ are integers, if $ord(\mathscr{T}) \geq 4\lambda + \mu + 2$ and there is no μ-stepped Σ-span $H', \eta', \mathscr{T}'$ which can be obtained from $H, \eta, \mathscr{T}$ by rearranging within λ of any $z \in A(H)$.

Let $H, \eta, \mathscr{T}$ be a Σ-span. A region r of H is an *eye* (of $H, \eta, \mathscr{T}$ in G) if

(i) there is a circuit C of H with $U(C) = bd(r)$

(ii) there are distinct vertices a, b, c, d of C such that a, b, c, d occur in $V(C)$ in that order, and $\{a, b, c, d\}$ is free with respect to $\mathscr{T}$

(iii) there are two disjoint $\eta(H)$-paths P, Q in G with ends $\eta(a), \eta(c)$ and $\eta(b), \eta(d)$ respectively.

Now let $r_1, ..., r_\kappa$ be eyes, with corresponding $\eta(H)$-paths P_i, Q_i $(1 \leq i \leq \kappa)$. We say that $r_1, ..., r_\kappa$ are *independent eyes* if the paths P_i, Q_i can be chosen so that

(i) for $1 \leq i < j \leq \kappa$, $d(r_i, r_j) = ord(\mathscr{T})$, where d is the metric of $\mathscr{T}$

(ii) for $1 \leq i < j \leq \kappa$, $P_i \cup Q_i$ is disjoint from $P_j \cup Q_j$.

Finally we can state our main result.

(1.1) *For any surface Σ with $bd(\Sigma) = \emptyset$, and any integers $\kappa, \phi, \mu \geq 0$, there are integers $\theta, \lambda, \rho \geq 0$ such that the following holds. Let $\mathscr{T}^*$ be a tangle in a graph G, such that some Σ-span of order $\geq \theta$ is (λ, μ)-flat. Then either*

(i) *there is a Σ-span of order $\geq \phi$ with $> \kappa$ independent eyes, or*

(ii) *there is a Σ'-span of order $\geq \phi$, where Σ' is a surface obtained by adding a crosscap to Σ, or*

(iii) *there is a $\mathscr{T}^*$-central segregation of G of type (ρ, κ) with an arrangement in Σ.*

Result (1.1) is a consequence of the following.

(1.2) *For any surface Σ with $bd(\Sigma) = \emptyset$, and any integers $k \geq 0$, $\phi \geq 8$, $\mu \geq 12$ and even, $\lambda_{k+1} \geq 4$, and $\theta_{k+1} \geq 4\lambda_{k+1} + \mu + 2$, there are integers $\rho_k, \lambda_k \geq 0$ and $\theta_k \geq 4\lambda_k + \mu + 2$ such that the following holds. Let $\mathscr{T}^*$ be a tangle in a graph G, and suppose that there is a (λ_k, μ)-flat Σ-span of order $\geq \theta_k$ with k eyes mutually at distance $\geq \theta_k$. Then either*

(i) *there is a (λ_{k+1}, μ)-flat Σ-span of order θ_{k+1} with $k+1$ eyes, mutually at distance θ_{k+1}, or*

[p. 116] (ii) *there is a Σ'-span of order ϕ, where Σ' is a surface obtained by adding a crosscap to Σ, or*

(iii) *there is a $\mathscr{T}^*$-central segregation of G of type (ρ_k, k) with an arrangement in Σ.*

Proof of (1.1), *assuming* (1.2). Let $\Sigma, \kappa, \phi, \mu$ be as in (1.1). We may assume that $\mu \geq 12$ and is even, for every (λ, μ)-flat Σ-span is also (λ, μ')-flat for all $\mu' \geq \mu$ (see (5.1)). We may also assume that $\phi \geq 8$, for if (1.1)

holds for $\phi \geq 8$ then it holds for all ϕ. Define $\theta_{\not\kappa+1} = \max(\phi, \mu+16)$ and $\lambda_{\not\kappa+1} = 4$;[F1] and for $k = \kappa, \kappa-1, \ldots, 0$, define $\theta_k, \lambda_k, \rho_k$ inductively by (1.2). Let $\theta = \theta_0$, $\lambda = \lambda_0$ and $\rho = \max(\rho_0, \ldots, \rho_\kappa)$. We claim that θ, λ, ρ satisfy (1.1). For let $G, \mathscr{T}^*$ be as in (1.1); then there is a (λ_0, μ)-flat Σ-span of order $\geq \theta_0$. Choose $k \geq 0$ with $k \leq \kappa+1$ maximum such that there is a (λ_k, μ)-flat Σ-span of order $\geq \theta_k$ (with order θ_k if $k \geq 1$), with k eyes mutually at distance $\geq \theta_k$. Suppose first that $k \leq \kappa$. Since (1.2)(i) does not hold, it follows from (1.2) that either (1.2)(ii) or (1.2)(iii) holds, and hence either (1.1)(ii) or (1.1)(iii) holds (because a segregation of type (ρ_k, κ) is also of type (ρ, κ)). We may therefore assume that $k = \kappa+1$. Let $H, \eta, \mathscr{T}$ be a $(\lambda_{\not\kappa+1}, \mu)$-flat Σ-span of order $\theta_{\kappa+1}$,[F2] with $\kappa+1$ eyes $r_1, \ldots, r_{\kappa+1}$ mutually at distance $\theta_{\kappa+1}$. Since $H, \eta, \mathscr{T}$ is $(4, \mu)$-flat, and hence $(0, \mu)$-flat (see (5.1)), there is no $\eta(H)$-path in G with ends $\eta(u), \eta(v)$, where $u \in V(H) \cap \bar{r}_i$, $v \in V(H) \cap \bar{r}_j$, for $i \neq j$, since $d(u, v) \geq \mu$ because

$$\mu + 2 \leq \theta_{\kappa+1} = d(r_i, r_j) \leq d(u, r_i) + d(u, v) + d(v, r_j) \leq d(u, v) + 2.$$

Hence $r_1, \ldots, r_{\kappa+1}$ are independent eyes, and so (1.1)(i) holds. □

2. Tangles And Transactions

To prove (1.2) we need at some stage to prove that a certain segregation is $\mathscr{T}^*$-central, and to do so we shall use the following lemma.

(2.1) *Let $\mathscr{T}$ be a tangle in a graph G, let $\rho \geq 1$ with $ord(\mathscr{T}) \geq 5\rho + 2$, and let $\mathscr{S}$ be a segregation of G of type (ρ, κ) for some κ. Suppose that for every $(C, \Omega) \in \mathscr{S}$ there is no $(A, B) \in \mathscr{T}$ of order $\leq 2\rho + 1$ with $B \subseteq C$. Then $\mathscr{S}$ is $\mathscr{T}$-central.*

Proof. Let $(C, \Omega) \in \mathscr{S}$, and suppose, for a contradiction, that $(A, B) \in \mathscr{T}$ and $B \subseteq C$. If $|\Omega| \geq 4$ then (C, Ω) is a ρ-vortex since $\mathscr{S}$ is of type (ρ, κ). If $|\bar{\Omega}| \leq 3$ then (C, Ω) has no transaction of cardinality ≥ 2, and again it is a ρ-vortex since $\rho \geq 1$. We may assume that $\bar{\Omega} \neq \emptyset$.[F3] Hence in either case, from [1, Theorem (8.1)] we may enumerate $\bar{\Omega} = \{t_1, \ldots, t_n\}$ in order so that there are subgraphs $X_1, \ldots, X_n$ of C satisfying (1)–(4) below:

[F1] **Misprint.** No k is bound here, and the induction that follows starts from $\theta_{\kappa+1}$ and $\lambda_{\kappa+1}$.

[F2] **Misprint.** With $k = \kappa+1$, λ_{k+1} would be $\lambda_{\kappa+2}$, but the span given by the choice of k is (λ_k, μ)-flat, that is, $(\lambda_{\kappa+1}, \mu)$-flat, and the next sentence uses $\lambda_{\kappa+1} = 4$.

[F3] **Proof gap.** A gap in the proof: if $\bar{\Omega} = \emptyset$ and C is not null, Graph Minors IX, (8.1) does not apply, since a linear decomposition needs $n = |\bar{\Omega}| \geq 1$ (see the note to (8.1) in Graph Minors IX). In that case C shares no vertex with the rest of G, so there is a separation of G of order $0 \leq 2\rho + 1$ whose second part is C. The hypothesis of (2.1), with the tangle axioms, then rules out $(A, B) \in \mathscr{T}$ with $B \subseteq C$ directly. So (2.1) holds as printed.

[p. 117] (1) $X_1 \cup \cdots \cup X_n = C$.

(2) For $1 \le i < j \le n$, $E(X_i \cap X_j) = \emptyset$ and $|V(X_i \cap X_j)| \le \rho$.

(3) For $1 \le i \le j \le k \le n$, $X_i \cap X_k \subseteq X_j$.

(4) For $1 \le i \le n$, $t_i \in V(X_i)$.

For any subgraph X of G, let $\hat{X}$ be the unique minimal subgraph so that $(X, \hat{X})$ is a separation of G, and let $N(X) = V(X \cap \hat{X})$. Thus, $N(X)$ is the set of all $v \in V(X)$ incident with an edge of G not in $E(X)$. We define X_0 and X_{n+1} to be the null graph.

(5) *For* $1 \le i \le n$, $N(X_i) \subseteq V(X_i \cap X_{i-1}) \cup V(X_i \cap X_{i+1}) \cup \{t_i\}$, *and consequently* $|N(X_i)| \le 2\rho + 1$.

Subproof. The second claim follows from the first and (2). To prove the first claim, let $v \in N(X_i)$, incident with an edge e of G not in $E(X_i)$. If $v \in V(X_j)$ for some j with $1 \le j < i$, then by (3) $v \in V(X_{i-1})$ and hence $v \in V(X_i \cap X_{i-1})$. If $v \in V(X_j)$ for some j with $i < j \le n$ then similarly $v \in V(X_i \cap X_{i+1})$. If $v \notin V(X_j)$ for all $j \ne i$ with $1 \le j \le n$, then $e \notin E(X_j)$ for $1 \le j \le n$, and so $e \notin E(C)$; hence $v \in N(C) \subseteq \bar{\Omega}$ and so $v = t_j$ for some j. Hence $v \in V(X_j)$ by (4), and so $j = i$ and $v = t_i$ as required.

Let us say that $X \subseteq G$ is *small* if $(X, \hat{X}) \in \mathscr{T}$.

(6) $X_1, \ldots, X_n$ *are small.*

Subproof. Let $1 \le i \le n$. By (5), $|N(X_i)| \le 2\rho + 1$, and so one of $(X_i, \hat{X}_i), (\hat{X}_i, X_i) \in \mathscr{T}$, since $ord(\mathscr{T}) \ge 2\rho + 2$. But from the hypothesis, $(\hat{X}_i, X_i) \notin \mathscr{T}$, and so X_i is small as required.

(7) *If* $Z \subseteq X \cup Y \subseteq G$ *and* X, Y *are small and* $|N(Z)| < ord(\mathscr{T})$, *then* Z *is small.*

Subproof. Since $(X, \hat{X}), (Y, \hat{Y}) \in \mathscr{T}$ it follows that $(\hat{Z}, Z) \notin \mathscr{T}$ by the second tangle axiom, since

$$G = Z \cup \hat{Z} \subseteq X \cup Y \cup \hat{Z}.$$

Since $(Z, \hat{Z})$ has order $< ord(\mathscr{T})$ it follows from the first axiom that $(Z, \hat{Z}) \in \mathscr{T}$ and hence Z is small. This proves (7).

We recall that $(A, B) \in \mathscr{T}$ and $B \subseteq C = X_1 \cup \cdots \cup X_n$. Let $R = V(A \cap B)$. For $1 \le i \le n$, let [p. 118]

$$P_i = \bigcup (X_j : 1 \le j \le i)$$

$$Q_i = \bigcup (X_j : i \le j \le n),$$

and let P_0 and Q_{n+1} be the null graph.

(8) *For* $0 \le i \le n$, *if* $|V(P_i) \cap R| < ord(\mathscr{T}) - \rho$ *then* $P_i \cap B$ *is small.*

Subproof. The result holds if $i = 0$, and we assume that $i \geq 1$ and proceed by induction on i. Since $V(P_{i-1}) \cap R \subseteq V(P_i) \cap R$ and hence $|V(P_{i-1}) \cap R| < ord(\mathscr{T}) - \rho$, it follows from the inductive hypothesis that $P_{i-1} \cap B$ is small. Since X_i is small by (6), and $P_i \cap B \subseteq (P_{i-1} \cap B) \cup X_i$, it suffices by (7) to show that $|N(P_i \cup B)| < ord(\mathscr{T})$.[F4] To show this we shall prove that

$$N(P_i \cap B) \subseteq (V(P_i) \cap R) \cup V(X_i \cap X_{i+1});$$

for since $|V(P_i) \cap R| < ord(\mathscr{T}) - \rho$ and $|V(X_i \cap X_{i+1})| \leq \rho$, the desired inequality is a consequence. Thus, let $v \in N(P_i \cap B)$, incident with an edge e of G not in $E(P_i \cap B)$. If $e \notin E(B)$ then $v \in N(B) \subseteq R$ as required. If $e \in E(B)$, then since $e \notin E(P_i \cap B)$ and $B \subseteq C$ it follows that $e \in E(X_j)$ for some j with $i < j \leq n$. Hence $v \in V(X_j)$, and since $v \in V(P_i)$ it follows from (3) that $v \in V(X_i \cap X_{i+1})$. This proves that

$$N(P_i \cap B) \subseteq V(P_i \cap R) \cup V(X_i \cap X_{i+1})$$

as claimed, and (8) follows.

Now choose j with $0 \leq j \leq n$, maximum such that $|V(P_j) \cap R| < ord(\mathscr{T}) - 2\rho - 1$. (This is possible, since P_0 is null.)

(9) *$j < n$, and $(P_j \cap B) \cup X_{j+1}$ is small, and $|P_{j+1} \cap R| \geq ord(\mathscr{T}) - 2\rho - 1$.*

Subproof. Now $P_j \cap B$ is small by (8). Since $P_n \cap B$ is not small because $(A, B) \in \mathscr{T}$, it follows that $j < n$. Now

$$N((P_j \cap B) \cup X_{j+1}) \subseteq N(P_j \cap B) \cup N(X_{j+1})$$

and so $|N((P_j \cap B) \cup X_{j+1})| < ord(\mathscr{T})$, since $|N(P_j \cap B)| < ord(\mathscr{T}) - 2\rho - 1$ by hypothesis and $|N(X_{j+1})| \leq 2\rho + 1$ by (5). From (7) we deduce that $(P_j \cap B) \cup X_{j+1}$ is small. Finally, from the maximality of j it follows that $|P_{j+1} \cap R| \geq ord(\mathscr{T}) - 2\rho - 1$. This proves (9).

Choose k with $1 \leq k \leq n + 1$, minimum such that $|V(Q_k) \cap R| < ord(\mathscr{T}) - 2\rho - 1$. Then, similarly,

[p. 119] (10) *$k > 1$, and $(Q_k \cap B) \cup X_{k-1}$ is small, and $|Q_{k-1} \cap R| \geq ord(\mathscr{T}) - 2\rho - 1$.*

Suppose that $k \leq j + 3$. Then

$$B \subseteq ((P_j \cap B) \cup X_{j+1}) \cup ((Q_k \cap B) \cup X_{k-1})$$

[F4] **Misprint.** Claim (8) is about $P_i \cap B$, and the next display bounds $N(P_i \cap B)$.

contrary to (7). Thus $k \geq j+4$, and hence $P_{j+1} \cap Q_{k-1} \subseteq X_{j+1} \cap X_{j+2}$ and so $|V(P_{j+1} \cap Q_{k-1})| \leq \rho$. But by (9) and (10),

$$\begin{aligned} 2(ord(\mathscr{T}) - 2\rho - 1) &\leq |P_{j+1} \cap R| + |Q_{k-1} \cap R| \\ &= |(P_{j+1} \cup Q_{k-1}) \cap R| + |(P_{j+1} \cap Q_{k-1}) \cap R| \\ &\leq |R| + \rho < ord(\mathscr{T}) + \rho \end{aligned}$$

since $R = V(A \cup\!\!\!\cap B)$[F5] and $(A, B) \in \mathscr{T}$. Consequently, $ord(\mathscr{T}) < 5\rho + 2$, a contradiction. Hence there is no such (A, B), as required. □

3. Rigidity

In this section we prove a lemma that will permit us to modify a rigid drawing and infer that the result remains rigid. First, we observe the following.

(3.1) *Let H be a rigid drawing in a surface Σ, and let $\mathscr{T}$ be a tangle in H of order ≥ 3. Let $\Delta \subseteq \Sigma$ be a closed disc with $bd(\Delta) \cap U(H) \subseteq V(H)$ and $|bd(\Delta) \cap V(H)| \leq 2$. Then $(H \cap \Delta, H \cap \overline{\Sigma - \Delta}) \in \mathscr{T}$ if and only if Δ is a dial for $bd(\Delta)$.*

Proof. Since H is rigid, there is a dial Δ' for $bd(\Delta)$. Let $A = H \cap \Delta'$, $B = H \cap \overline{\Sigma - \Delta'}$. Then A has no circuit, and so $(B, A) \notin \mathscr{T}$ by [2, Theorem (2.10)]. Since (A, B) has order ≤ 2, and $ord(\mathscr{T}) \geq 3$, it follows that $(A, B) \in \mathscr{T}$. Now either $\Delta = \Delta'$ or $\Delta = \overline{\Sigma - \Delta'}$, and in either case the result follows. □

The main result of this section is the following cumbersome lemma, which we shall have several occasions to use.

(3.2) *Let Σ be a surface with $bd(\Sigma) = \emptyset$, and let $\Sigma_1, \Sigma_2 \subseteq \Sigma$ be surfaces with $\Sigma_1 \cup \Sigma_2 = \Sigma$ and $\Sigma_1 \cap \Sigma_2 = bd(\Sigma_1) = bd(\Sigma_2)$. Let H be a drawing in Σ with $\Sigma_1 \cap \Sigma_2 \subseteq U(H)$, and let $X \subseteq V(H) \cap \Sigma_1 \cap \Sigma_2$. Let $\hat{\Sigma}_1$ be a surface with $\Sigma_1 \subseteq \hat{\Sigma}_1$ and $bd(\hat{\Sigma}_1) = \emptyset$, such that each 0-arc in $bd(\Sigma_1)$ bounds an open disc in $\hat{\Sigma}_1$ disjoint from Σ_1 (thus, $\hat{\Sigma}_1$ is obtained from Σ_1 by "pasting" a disc onto each 0-arc in $bd(\Sigma_1)$). Suppose that*

[p. 120] (i) *if r is a region of H in Σ and $r \subseteq \Sigma_1$, there is at most one component F of $\Sigma_1 \cap \Sigma_2$ with $\bar{r} \cap F \neq \emptyset$*

(ii) *for every 0-arc $F \subseteq \hat{\Sigma}_1$, with $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| \leq 2$, there is a dial $\Delta \subseteq \hat{\Sigma}_1$ for $F, H \cap \Sigma_1$ with $|X \cap \Delta| \leq |F \cap V(H)|$*

(iii) *for every 0-arc $F \subseteq \Sigma_2$ with $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| \leq 2$, there is a dial $\Delta \subseteq \Sigma_2$ for F, H*

[F5] **Misprint.** Here R is defined as $V(A \cap B)$.

(iv) *if $F \subseteq \Sigma_2$ is a line with ends $u, v \in \Sigma_1 \cap \Sigma_2 \cap V(H)$ and is otherwise disjoint from $U(H)$, and P is a path of $H \cap \Sigma_1 \cap \Sigma_2$ with ends u, v and with $|V(P) \cap X| \leq 2$, then the O-arc $U(P) \cup F$ bounds an open disc in Σ_2 disjoint from $U(H)$.*

Then H is rigid in Σ.

Proof. Let $F \subseteq \Sigma$ be an O-arc with $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| \leq 2$. We shall show that Σ includes a dial for F, H. From (iii) we may assume that $F \not\subseteq \Sigma_2$. Suppose that $F \subseteq \Sigma_1$. From (ii), there is a dial $\Delta \subseteq \hat{\Sigma}_1$ for $F, H \cap \Sigma_1$. Every component of $\hat{\Sigma}_1 - \Sigma_1$ is bounded by an O-arc in $\Sigma_1 \cap \Sigma_2 \subseteq U(H)$, and hence Δ includes no such component, since $(H \cap \Sigma_1) \cap \Delta$ has no circuit. Since $bd(\Delta) \subseteq \Sigma_1$, it follows that $\Delta \subseteq \Sigma_1 \subseteq \Sigma$, and hence Δ is a dial in Σ for F, H as required. We may assume then that $F \not\subseteq \Sigma_1$.

Since $F \not\subseteq \Sigma_1$ and $F \not\subseteq \Sigma_2$, it follows that $|F \cap \Sigma_1 \cap \Sigma_2| \geq 2$. Since $F \cap U(H) \subseteq V(H)$ and $\Sigma_1 \cap \Sigma_2 \subseteq U(H)$ and $|F \cap V(H)| \leq 2$, we deduce that $F \cap \Sigma_1 \cap \Sigma_2 = \{u, v\} \subseteq V(H)$ say, and $F = F_1 \cup F_2$ where F_1, F_2 are lines with ends u, v and $F_i \subseteq \Sigma_i$ $(i = 1, 2)$. For $i = 1, 2$, let r_i be the region of H in Σ with $F_i \subseteq r_i \cup \{u, v\}$. Then $r_i \subseteq \Sigma_i$ $(i = 1, 2)$. From (i), u and v lie in the same component C of $\Sigma_1 \cap \Sigma_2$. Let Λ be the component of $\hat{\Sigma}_1 - \Sigma_1$ bounded by $U(C)$. By adding a line joining u, v within $\Lambda \cup \{u, v\}$ we extend F_1 to an O-arc in $\hat{\Sigma}_1$, and thereby deduce from (ii) that there is a path P of C between u and v such that $|V(P) \cap X| \leq 2$, and $F_1 \cup U(P)$ bounds an open disc in Σ_1 included in r_1. From (iv), $F_2 \cup U(P)$ bounds an open disc in Σ_2 included in r_2. It follows that the union of the closures of these open discs is a dial in Σ for F, H, as required. □

If $\mathscr{T}$ is a tangle of order θ in a graph H, and $1 \leq \theta' \leq \theta$, the set $\mathscr{T}'$ of all members of $\mathscr{T}$ of order $< \theta'$ is a tangle in H of order θ', and we call it the *θ'-truncation* of $\mathscr{T}$.

We apply (3.2) to deduce the following.

(3.3) *Let Σ be a surface and let $\phi \geq 3$ be an integer. Let $\mathscr{T}^*$ be a tangle in a graph G, and let $H, \eta, \mathscr{T}$ be a Σ-span. Let $\Lambda \subseteq \Sigma$ be a region of H such that $H \cap bd(\Lambda)$ is a circuit, and let $a_1, \ldots, a_\phi, b_1, \ldots, b_\phi$ be distinct vertices of this circuit, in order, such that $\{a_1, \ldots, a_\phi, b_1, \ldots, b_\phi\}$ is free with respect to $\mathscr{T}$. For $1 \leq i \leq \phi$ let P_i be an $\eta(H)$-path with ends $\eta(a_i), \eta(b_i)$, such that* [p. 121] *$P_1, \ldots, P_\phi$ are mutually disjoint. Then there is a Σ'-span of order ϕ, where Σ' is the surface obtained by adding a crosscap to Σ.*

Proof. Let $\Sigma_1 = \Sigma - \Lambda$. We may assume that $\Sigma' = \Sigma_1 \cup \Sigma_2$, where Σ_2 is homeomorphic to a Möbius band and $\Sigma_1 \cap \Sigma_2 = bd(\Sigma_1) = bd(\Sigma_2)$. We may assume also that η is the identity, and $P_1, \ldots, P_\phi$ are drawings in Σ_2, so that $(H \cap bd(\Lambda)) \cup P_1 \cup \cdots \cup P_\phi$ is a drawing in Σ_2. Let $H' = H \cup P_1 \cup \cdots \cup P_\phi$;

then H' is a drawing in Σ'. Let $\mathscr{T}''$ be the tangle in H' of order θ induced by $\mathscr{T}$, and let $\mathscr{T}'$ be the ϕ-truncation of $\mathscr{T}''$. We claim that $H', \mathscr{T}'$ is a Σ'-span of order ϕ. To show this, we must show

(1) *The following statements (i)–(iii) hold:*

(i) H' *is a rigid drawing in* Σ'

(ii) $\mathscr{T}'$ *is a respectful tangle in* H' *of order* ϕ

(iii) $\phi \le ord(\mathscr{T}^*)$*, and* $(A \cap H', B \cap H') \in \mathscr{T}'$ *for every* $(A, B) \in \mathscr{T}^*$ *of order* $< \phi$.

Let $\mathscr{T}_1$ be the ϕ-truncation of $\mathscr{T}$. Then $\mathscr{T}'$ is the tangle in H' induced by $\mathscr{T}_1$. We claim

(2) *Every* $X \subseteq \{a_1, \dots, a_\phi, b_1, \dots, b_\phi\}$ *with* $|X| \le \phi$ *is free with respect to* $\mathscr{T}_1$.

Subproof. If X is not free, then since $|X| \le \phi$ and $\mathscr{T}_1$ has order ϕ, there exists $(A, B) \in \mathscr{T}_1$ of order $< |X|$ with $X \subseteq V(A)$. Then $(A, B) \in \mathscr{T}$, and so X is not free with respect to $\mathscr{T}$, contrary to [2, Theorem (12.2)]. This proves (2).

From (2) and [4, Theorem (5.2)] we deduce that $\mathscr{T}'$ is respectful, so (1)(ii) holds. Next we show that (1)(iii) holds. For $ord(\mathscr{T}) \le ord(\mathscr{T}^*)$ since $H, \eta, \mathscr{T}$ is a Σ-span, and $2\phi \le ord(\mathscr{T})$ since $\{a_1, \dots, a_\phi, b_1, \dots, b_\phi\}$ is free with respect to $\mathscr{T}$. Thus $\phi \le 2\phi \le ord(\mathscr{T}^*)$. If $(A, B) \in \mathscr{T}^*$ has order $< \phi$, then $(A \cap H', B \cap H')$ is a separation of H' of order $< \phi$, and

$$((A \cap H') \cap H, (B \cap H') \cap H) = (A \cap H, B \cap H) \in \mathscr{T}$$

because $H, \eta, \mathscr{T}$ is a Σ-span. Hence $(A \cap H', B \cap H') \in \mathscr{T}''$ by definition of $\mathscr{T}''$, and hence $(A \cap H', B \cap H') \in \mathscr{T}'$ as required. Thus (1)(iii) holds.

To prove (1) it remains to show that H' is rigid, and to show this we shall use (3.2). Let $X = \{a_1, \dots, a_\phi, b_1, \dots, b_\phi\}$.

(3) *For every O-arc* $F \subseteq \Sigma$ *with* $F \cap U(H') \subseteq V(H')$ *and* $|F \cap V(H')| \le 2$*, there is a dial* $\Delta \subseteq \Sigma$ *for* $F, H' \cap \Sigma_1$ *with* $|X \cap \Delta| \le |F \cap V(H')|$.

[p. 122] *Subproof.* Now $H' \cap \Sigma_1 = H$, and since H is rigid there is a dial $\Delta \subseteq \Sigma$ for F, H. By (3.1) it follows that $(H \cap \Delta, H \cap \overline{\Sigma - \Delta}) \in \mathscr{T}$. Since X is free with respect to $\mathscr{T}$, we deduce that

$$\begin{aligned} |X \cap \Delta| &= |X \cap V(H \cap \Delta)| \le |V((H \cap \Delta) \cap (H \cap \overline{\Sigma - \Delta}))| \\ &= |V(H) \cap F| = |F \cap V(H')|. \end{aligned}$$

This proves (3).

(4) *For every O-arc* $F \subseteq \Sigma_2$ *with* $F \cap U(H')V(H')$ *and* $|F \cap V(H')| \le 2$ *there is a dial* $\Delta \subseteq \Sigma_2$ *for* F, H'.

This is easy to see, because since $\phi \geq 3$ and $|F \cap V(H')| \leq 2$, there exists i such that $F \cap U(P_i) = \emptyset$, and hence F bounds a closed disc in Σ_2.

(5) *If $F \subseteq \Sigma_2$ is a line with ends $u, v \in \Sigma_1 \cap \Sigma_2 \cap V(H')$ and is otherwise disjoint from $U(H')$, and P is a path of $H' \cap \Sigma_1 \cap \Sigma_2$ with ends u, v and with $|V(P) \cap X| \leq 2$, then the O-arc $U(P) \cup F$ bounds an open disc in Σ_2 disjoint from $U(H)$.*

Subproof. Since $|U(P) \cap X| \leq 2$ and $F \cap U(H') \subseteq U(P)$, it follows that $U(P) \cup F$ meets at most two of $P_1, \ldots, P_\phi$, and hence it bounds a disc in Σ_2. Again, the claim follows easily.

From (3), (4), and (5) and the fact that $\Sigma_1 \cap \Sigma_2$ has only one component, it follows from (3.2) that H' is rigid in Σ'. This proves (1) and completes the proof of the theorem. □

A second application of (3.2) is the following.

(3.4) *Let Σ be a surface and let $k \geq 3$ be an integer. Let $\mathscr{T}^*$ be a tangle in a graph G, and let $H, \mathscr{T}$ be a Σ-span of order θ. Let $\Lambda \subseteq \Sigma$ be a region of H such that $H \cap bd(\Lambda)$ is a circuit, and let $v_1, \ldots, v_{2k}$ be distinct vertices of this circuit, in order, such that $\{v_1, \ldots, v_{2k}\}$ is free with respect to $\mathscr{T}$. Let $H' \subseteq G$ be a drawing in Σ such that $H' \cap (\Sigma - \Lambda) = H$ and $H' = H \cup P_1 \cup \cdots \cup P_{k-1}$, where P_i is a path with ends v_i, v_{2k-i} for $1 \leq i \leq k-1$, the paths $P_1, \ldots, P_{k-1}$ are mutually disjoint, and $U(P_i) \cap U(H) = \{v_i, v_{2k-i}\}$ for $1 \leq i \leq k-1$.*[F6] *Let $\mathscr{T}'$ be a tangle in H' induced by $\mathscr{T}$; then $H', \mathscr{T}'$ is a Σ-span of order θ. Moreover, let $r_1, r_2 \subseteq \Lambda$ be the regions of H' incident with v_k, v_{2k} respectively; then $d'(r_1, r_2) \geq 2k-6$, and $d'(v_k, v_{2k}) \geq 2k-8$, where d' is the metric of $\mathscr{T}'$.*

Proof. To show that $H', \mathscr{T}'$ is a Σ-span of order θ, we must show

(1) *The following statements (i)–(iii) hold:*

(i) *H' is a rigid drawing in Σ*

(ii) *$\mathscr{T}'$ is a respectful tangle of order θ, and*

[p. 123] (iii) *$\theta \leq ord(\mathscr{T}')$, and $(A \cap H', B \cap H') \in \mathscr{T}'$ for every $(A, B) \in \mathscr{T}^*$ of order $< \theta$.*

(1)(ii) is clear, and (1)(iii) is proved in the same way as in the proof of (3.3). It remains to show (1)(i). Let $\Sigma_1 = \Sigma - \Lambda$, $\Sigma_2 = \bar{\Lambda}$ and $X = \{v_1, \ldots, v_{2k}\}$. We shall verify the hypotheses of (3.2). Now (3.2)(i) is trivial since $\Sigma_1 \cap \Sigma_2$ is connected; and (3.2)(ii) is proved as in the proof of (3.3). For (3.2)(iii), let $F \subseteq \Sigma_2$ be an O-arc with $F \cap U(H') \subseteq V(H')$ and $|F \cap V(H')| \leq$

[F6] **Statement mistake.** The proof uses both added conditions: "$H' \cap \Sigma_2$ is the union of $H \cap bd(\Lambda)$ and the disjoint paths $P_1, \ldots, P_{k-1}$". Without them the bound on $d'(r_1, r_2)$ fails. If each P_i runs along the circuit $H \cap bd(\Lambda)$, then $H' = H$ and $r_1 = r_2 = \Lambda$. If every P_i has two edges through one vertex x in Λ, then r_1 and r_2 are both incident with x, and $d'(r_1, r_2) \leq 2$. The uses in (7.10) and (7.11) satisfy both conditions.

2; then the closed disc in Σ_2 bounded by F is a dial for F because $H' \cap \Sigma_2$ is the union of $H \cap bd(\Lambda)$ and the disjoint paths $P_1, \ldots, P_{k-1}$. For (3.2)(iv), let u, v, F, P be as in (3.2)(iv). If $F \subseteq r_1 \cup \{u, v\}$ or $F \subseteq r_2 \cup \{u, v\}$, then (3.2)(iv) is satisfied, and so we may assume that $F \subseteq r_3 \cup \{u, v\}$ for some region $r_3 \neq r_1, r_2$ of H' with $r_3 \subseteq \Lambda$. Let the four vertices in X incident with r_3 be $v_i, v_{i+1}, v_{2k-i-1}, v_{2k-i}$, where $1 \leq i \leq k-2$. Now there is a path of $H \cap \Sigma$ between u and v, passing through at most two vertices in X; and so u, v both lie in the same component of $H \cap bd(r_3)$ (because of the vertices $v_k, v_{2k} \in X$) and again (3.2)(iv) is satisfied. We give the details. First, for $1 \leq j \leq k-1$ we show that $U(P_j) \subseteq \Lambda \cup \{v_j, v_{2k-j}\}$. Let e be an edge of P_j. Since $U(P_j) \cap U(H) = \{v_j, v_{2k-j}\}$ and these are vertices, $e \cap U(H) = \emptyset$, and so $e \cap bd(\Lambda) = \emptyset$. Since e is connected, either $e \subseteq \Lambda$ or $e \subseteq \Sigma - \bar{\Lambda}$; and the second is impossible, since then e would be an edge of $H' \cap (\Sigma - \Lambda) = H$. Thus every edge of P_j is a subset of Λ, and so every vertex of P_j is in $\bar{\Lambda}$. Every vertex of P_j other than v_j, v_{2k-j} is not in $U(H)$, and hence not in $bd(\Lambda)$, and so it is in Λ. This proves that $U(P_j) \subseteq \Lambda \cup \{v_j, v_{2k-j}\}$. Since $U(H) \cap \Lambda = \emptyset$, it follows that $\Sigma_2 \cap U(H') = bd(\Lambda) \cup U(P_1) \cup \cdots \cup U(P_{k-1})$, and that $H' \cap bd(\Lambda) = H \cap bd(\Lambda)$.

Now let u, v, F, P be as in (3.2)(iv). We show first that for $1 \leq j \leq k-1$, P does not contain both v_j and v_{2k-j}. The circuit $H \cap bd(\Lambda)$ is the union of two paths with ends v_j, v_{2k-j}, and since $v_1, \ldots, v_{2k}$ are in order on this circuit, one of these paths contains v_k and the other contains v_{2k}. If $v_j, v_{2k-j} \in V(P)$, then since P is a path of this circuit, P includes one of these two paths, and so $|V(P) \cap X| \geq 3$, a contradiction.

Since F meets $U(H')$ only in u and v, the set $U(P) \cup F$ is an O-arc in the closed disc Σ_2, and so it bounds a closed disc $D \subseteq \Sigma_2$. Let $D^0 = D - (U(P) \cup F)$. Then D^0 is an open disc, and $D^0 \subseteq \Lambda$, since D^0 is open in Σ and $D^0 \subseteq \bar{\Lambda}$. We claim that $D^0 \cap U(H') = \emptyset$. For suppose that $x \in D^0 \cap U(H')$. Since $x \in \Lambda$, the first paragraph implies that $x \in U(P_j) - \{v_j, v_{2k-j}\}$ for some j. This set is connected, and it is disjoint from $U(P) \cup F$, since $U(P) \cup (F \cap U(H')) \subseteq bd(\Lambda)$. Since it meets D^0 and $\Sigma - D$ is open, it is a subset of D^0, and so $v_j, v_{2k-j} \in D$. But $v_j, v_{2k-j} \in bd(\Lambda)$, and so they are not in D^0. Hence they belong to $U(P) \cup F$, and therefore to $U(P)$, since $F \cap U(H') = \{u, v\} \subseteq U(P)$. Thus $v_j, v_{2k-j} \in V(P)$, contrary to what we showed above. This proves that $D^0 \cap U(H') = \emptyset$, and so (3.2)(iv) is satisfied.[F7]

[F7] **Missing proof.** The paper does not prove that the vertices of X incident with r_3 are $v_i, v_{i+1}, v_{2k-i-1}, v_{2k-i}$, and it does not say why (3.2)(iv) holds when $F \subseteq r_1 \cup \{u, v\}$ or $F \subseteq r_2 \cup \{u, v\}$. The argument above is the one used in the formalization.

By (3.2), H' is rigid. Hence $H', \mathscr{T}'$ is a Σ-span of order θ. Let d' be the metric of $\mathscr{T}'$, and let e_1, e_2 be edges of P_1, P_{k-1} respectively. Now $ord(\mathscr{T}') \geq 2k > 2(k-1)$, and so by [4, Theorem (6.1)], $d'(e_1, e_2) \geq 2k-2$. We give the details. By the first paragraph of the details for (3.2)(iv) above, $U(P_1), \ldots, U(P_{k-1})$ are mutually disjoint lines in the closed disc $\bar{\Lambda}$, and $U(P_j) \cap bd(\Lambda) = \{v_j, v_{2k-j}\}$ for $1 \leq j \leq k-1$. Let D_0 be the closed unit disc in the plane, and for $1 \leq j \leq k-1$ let C_j be the vertical chord of D_0 with ends $(\cos(\pi j/k), \pm\sin(\pi j/k))$. Thus $C_1, \ldots, C_{k-1}$ are disjoint, and they occur in this order from right to left.

We claim that there is a homeomorphism $\phi\colon \bar{\Lambda} \to D_0$ that maps v_i to $(\cos(\pi i/k), \sin(\pi i/k))$ for $1 \leq i \leq 2k$, and maps $U(P_j)$ onto C_j for $1 \leq j \leq k-1$. Since $v_1, \ldots, v_{2k}$ are in order on the O-arc $bd(\Lambda)$, there is a homeomorphism ϕ with the first property. Suppose that $\phi(U(P_j)) = C_j$ for $1 \leq j < n$, where $1 \leq n \leq k-1$. Let $D_1 = D_0$ if $n = 1$, and otherwise let D_1 be the part of D_0 on the left of C_{n-1}, including C_{n-1}. The line $L = \phi(U(P_n))$ is disjoint from C_{n-1}, and its ends are on the left of C_{n-1}, so $L \subseteq D_1$. Both L and C_n are lines in the closed disc D_1 with the same ends, meeting $bd(D_1)$ only in their ends. Each of them divides D_1 into two closed discs, bounded by the line and by the two arcs of $bd(D_1)$ between its ends. Since a homeomorphism between the boundaries of two closed discs extends to a homeomorphism between the discs, there is a homeomorphism of D_1 onto itself that fixes $bd(D_1)$ and maps L onto C_n. We extend it by the identity on $D_0 - D_1$, and compose ϕ with the result. The new ϕ still maps $U(P_j)$ onto C_j for $j < n$, since $C_j \subseteq (D_0 - D_1) \cup bd(D_1)$, and now also for $j = n$. The claim follows by induction on n.

The image under ϕ^{-1} of the part of $D_0 - bd(D_0)$ on the left of C_{k-1} is connected and open, it is disjoint from $U(H')$, and its boundary is included in $U(H')$. Hence it is a region of H' in Λ. It is incident with v_k, and so it is r_1, and it is incident with every edge of P_{k-1}. Similarly r_2 is the image of the part of $D_0 - bd(D_0)$ on the right of C_1, and it is incident with v_{2k} and with every edge of P_1.

Now let ψ be a homeomorphism of the plane with $\psi(D_0) \subseteq D_0$ that fixes every point of $C_1 \cup \cdots \cup C_{k-1}$ and maps every other point of $bd(D_0)$ into $D_0 - bd(D_0)$. (For instance, let $c = \cos(\pi/k)$, let $h(x) = x$ if $|x| \leq c$ and $h(x) = (x+c)/2$ if $x > c$ and $h(x) = (x-c)/2$ if $x < -c$, let $s(x) = 1 - \min(1/2, \prod_{j=1}^{k-1}(x - \cos(\pi j/k))^2)$, and let $\psi(x,y) = (h(x), s(x)y)$.) Let $\Delta = \phi^{-1}(\psi(D_0))$ and $F = bd(\Delta)$. Then $\Delta \subseteq \bar{\Lambda}$ is a closed disc, $\Delta \cap bd(\Lambda)$ is the set Y of the $2k-2$ ends of $P_1, \ldots, P_{k-1}$, and every other point of $U(P_1) \cup \cdots \cup U(P_{k-1})$ is in $\Delta - F$. (For $\psi(D_0) \cap bd(D_0)$ is the set of ends

of $C_1, \ldots, C_{k-1}$, and ψ fixes each C_j and maps $D_0 - bd(D_0)$ onto $\psi(D_0) - bd(\psi(D_0))$.) Since $U(H) \cap \bar{\Lambda} \subseteq bd(\Lambda)$, it follows that $H' \cap \Delta = P_1 \cup \cdots \cup P_{k-1}$, that $H' \cap \overline{\Sigma - \Delta} = H$, and that $F \cap U(H') = Y$, so F is H'-normal. The members of Y occur on F in the order $v_1, \ldots, v_{k-1}, v_{k+1}, \ldots, v_{2k-1}$, since ψ fixes them and maps $bd(D_0)$ onto $bd(\psi(D_0))$; and P_j has ends the jth and the $(2k-1-j)$th of them. Moreover H' and H are 2-cell, since they are rigid; $\mathscr{T}$ is a respectful tangle in H of order $\theta \geq 2k > 2k-2$, where $\theta \geq 2k$ since X is free with respect to $\mathscr{T}$; $\mathscr{T}'$ is the tangle in H' induced by $\mathscr{T}$; and $Y \subseteq X$ is free with respect to $\mathscr{T}$, as in the proof of (2) in the proof of (3.3). Since $2k-2 \geq 4$, [4, Theorem (6.1)] applies with $\Gamma = H'$, with $2k-2$ in place of θ, and with $\Gamma_1 = H$ and $\mathscr{T}_1 = \mathscr{T}$, and it gives $d'(e_1, e_2) \geq 2k-2$.[F8] Hence $d'(r_1, r_2) \geq 2k-6$, since $d'(r_1, e_{12}) \leq 2$ and $d'(r_2, e_{21}) \leq 2$; and $d'(v_{1k}, v_{22k}) \geq 2k-8$, since $d'(v_{1k}, e_{12}) \leq 3$ and $d'(v_{22k}, e_{21}) \leq 3$.[F9] □

Our third application of (3.2) is the following, which will be needed in a future paper.

(3.5) *Let Σ be a surface and let $\phi \geq 3$ be an integer. Let $\mathscr{T}^*$ be a tangle in a graph G, and let $H, \eta, \mathscr{T}$ be a Σ-span. Let $\Lambda_1, \Lambda_2 \subseteq \Sigma$ be distinct regions of H so that $H \cap bd(\Lambda_i)$ is a circuit C_i $(i = 1, 2)$, and no region of H is incident with a vertex in C_1 and with a vertex in C_2, and every H-normal line $F \subseteq \Sigma - (\Lambda_1 \cup \Lambda_2)$ with one end in $bd(\Lambda_1)$ and the other in $bd(\Lambda_2)$ satisfies $|F \cap V(H)| \geq \phi$.*[F10] *Let $a_1, \ldots, a_\phi$ be distinct vertices of C_1 in order, and let $b_1, \ldots, b_\phi$ be distinct vertices of C_2 in order (under some orientations of C_1 and C_2), such that $\{a_1, \ldots, a_\phi\}$ and $\{b_1, \ldots, b_\phi\}$ are free with respect to $\mathscr{T}$. For $1 \leq i \leq \phi$ let P_i be an $\eta(H)$-path with ends $\eta(a_i), \eta(b_i)$, such that $P_1, \ldots, P_\phi$ are mutually disjoint. Then there is a Σ'-span of order ϕ, where Σ' is a surface obtained by adding a handle to Σ.*

Proof. We may assume that η is the identity. Let $\Sigma_1 = \Sigma - (\Lambda_1 \cup \Lambda_2)$. Let Σ_2 be homeomorphic to a closed cylinder, with boundary $U(C_1) \cup U(C_2)$,

[F8] **Missing proof.** The paper does not give the disc to which [4, Theorem (6.1)] is applied, and does not check the hypotheses of that result. The argument above is the one used in the formalization.

[F9] **Misprint.** Here r_1 is incident with the edges of P_{k-1}, among them e_2, and r_2 with the edges of P_1, among them e_1. The statement of (3.4) bounds $d'(v_k, v_{2k})$.

[F10] **Statement mistake.** The proof uses Graph Minors XII, (5.1), which needs the added hypothesis. Without it, (3.5) is false. In the sphere, take C_1 and C_2 joined by a line that meets only 4 vertices of H, and take $\phi \geq 5$. After the handle is added, this line and a line along P_1 form an O-arc that does not separate the torus but meets H' in fewer than ϕ vertices, so no tangle of order ϕ in H' is respectful. With the hypothesis, (3.5) holds. At its only use, in Claim (4) in the proof of 8.1 of Graph Minors XVI, page 72, Λ_1 and Λ_2 come from cleared zones around two vertices that are far apart in the metric of the tangle, and this gives the hypothesis.

such that $\Sigma' = \Sigma_1 \cup \Sigma_2$ is a surface obtained from Σ by adding a handle, and there are ϕ disjoint lines in Σ_2 with ends a_i, b_i $(1 \le i \le \phi)$. We may therefore assume that $P_1, \dots, P_\phi$ are drawings in Σ_2 and hence in Σ'. Let $H' = H \cup P_1 \cup \cdots \cup P_\phi$, let $\mathscr{T}''$ be the tangle in H' of order θ induced by $\mathscr{T}$, and let $\mathscr{T}'$ be the ϕ-truncation of $\mathscr{T}''$. We claim that [p. 124] $H', \mathcal{T}'$ is a Σ'-span of order ϕ. To show this, it follows by an argument similar to that in the proof of (3.3), using [4, Theorem (5.1)] in place of [4, Theorem (5.2)], that $\mathcal{T}'$ is respectful and that it suffices to prove that H' is rigid. To show that H' is rigid we verify (3.2)(i)-(iv). Now (3.2)(i) holds by hypothesis. For (3.2)(ii), let $X = \{a_1, \dots, a_\phi, b_1, \dots, b_\phi\}$.

(1) *For every O-arc $F \subseteq \Sigma$ with $F \cap U(H') \subseteq V(H')$ and $|F \cap V(H')| \le 2$, there is a dial $\Delta \subseteq \Sigma$ for $F, H' \cap \Sigma_1$ with $|X \cap \Delta| \le |F \cap V(H')|$.*

Subproof. As in the proof of (3) in (3.3), there is a dial $\Delta \subseteq \Sigma$ for F, H, and $(H \cap \Delta, H \cap \overline{\Sigma - \Delta}) \in \mathcal{T}$. Since $|F \cap V(H')| \le 2$ and no region of H is incident with a vertex of C_1 and a vertex of C_2, we may therefore assume that $F \cap U(C_2) = \emptyset$. Since $\{b_1, \dots, b_\phi\}$ is free with respect to $\mathcal{T}$ and $\phi \ge 3$, it follows that $\{b_1, \dots, b_\phi\} \not\subseteq \Delta$, and so $\bar{\Lambda}_2 \cap \Delta = \emptyset$. Moreover, since $\{a_1, \dots, a_\phi\}$ is free with respect to $\mathcal{T}$, we deduce that

$$|\{a_1, \dots, a_\phi\} \cap \Delta| \le |F \cap V(H')|.$$

Consequently $|X \cap \Delta| \le |F \cap V(H')|$. This proves (1).

(3.2)(ii) follows, and (3.2)(iii) is easy to see. (3.2)(iv) is proved as in step (5) of (3.3). From (3.2) we therefore deduce that H' is rigid in Σ', as required. □

4. Rearrangement

In this section, we develop an assortment of lemmas about compressions and rearrangement. If $\mathcal{T}$ is a respectful tangle of order θ in a 2-cell drawing H in Σ, and K is a radial ~~graph~~ drawing of ~~G~~H,[F11] we define $ins(C)$, for each circuit C of K with $< 2\theta$ edges, to be the closed disc $\Delta \subseteq \Sigma$ bounded by $U(C)$ such that

$$(H \cap \Delta, H \cap \overline{\Sigma - \Delta}) \in \mathcal{T}.$$

(This exists since $\mathcal{T}$ is respectful.) For any closed walk W of K of length $< 2\theta$, we define $ins(W)$ to be the union of $U(K')$ and $ins(C)$ taken over all circuits C of K', where $K' \subseteq K$ is the subdrawing formed by the vertices

[F11] **Misprint.** No G appears here. The tangle and the separations are in H, and C must be a circuit of a radial drawing of H, as in (4.1), which next says "Let H_1, H_2 have radial drawings K_1, K_2".

and edges in W. We call ins the slope corresponding to $\mathcal{T}$. For tangles $\mathcal{T}'$, $\mathcal{T}''$, $\mathcal{T}_0$ etc. we shall consistently denote the corresponding slopes by ins', ins'', ins_0 etc. without further definition, and we use a similar convention for metrics (d', d'', d_0 etc.).

(4.1) *Let $\mathcal{T}$ be a respectful tangle of order θ in a 2-cell drawing H in a surface Σ. Let H_1, H_2 be 2-cell drawings in Σ, and for $i = 1, 2$ let $\mathcal{T}_i$ be a λ_i-compression of $\mathcal{T}$ in H_i. Let H_1, H_2 have radial drawings K_1, K_2. Then* [p. 125] *for every W which is a closed walk of a common subdrawing of K_1 and K_2, if $ins_1(W)$, $ins_2(W)$ are both defined then they are equal. In particular, if $H_1 = H_2$ and $\lambda_1 \geq \lambda_2$, then $\mathcal{T}_1 \subseteq \mathcal{T}_2$.*

Proof. Since $ins_i(W)$ is determined by $ins_i(C)$ for the circuits C of the subdrawing formed by the vertices and edges in W, it suffices to show that $ins_1(C) = ins_2(C)$ for every C with $|E(C)| < 2\min(\theta - \lambda_1, \theta - \lambda_2)$ which is a circuit of both K_1 and K_2. Choose $v \in V(C)$. By [3, Theorem (8.12)], there exists $\sigma \in \Sigma$ with $d(v, \sigma) = \theta$, and hence with $d_i(v, \sigma) \geq \theta - \lambda$ for $i = 1, 2$. Hence $\sigma \notin ins_1(C)$ and $\sigma \notin ins_2(C)$, and so $ins_1(C) = ins_2(C)$, since there is at most one closed disc in Σ bounded by $U(C)$ not containing σ.

In particular, let $H_1 = H_2$ and $\lambda_1 \geq \lambda_2$; and hence we may assume that $K_1 = K_2 = K$. Then $ins_1(C) = ins_2(C)$ for every circuit C of K with $< 2(\theta - \lambda_1)$ edges, and hence $\mathcal{T}_1 \subseteq \mathcal{T}_2$ by [3, Theorem (6.3)]. □

(4.2) *Let H, H' be 2-cell drawings in a surface Σ, and let $Z \subseteq \Sigma$ include every atom of H which is not in $A(H')$. (Hence Z includes every atom of H' not in $A(H)$.) Let $\mathcal{T}, \mathcal{T}'$ be respectful tangles in H, H' respectively, such that, for some $\lambda \geq 0$, one of $\mathcal{T}, \mathcal{T}'$ is a λ-compression of the other. Then for all $x, y \in \Sigma$, if $d(x, y) < d(x, \sigma)$ for all $\sigma \in Z$, and $d(x, y) < ord(\mathcal{T})$, then $d'(x, y) \leq d(x, y)$.*

Proof. Let K be a radial drawing of H. Since $d(x, y) < ord(\mathcal{T})$, there is a closed walk W of K of length $2d(x, y)$ such that $ins(W) \cap a$, $ins(W) \cap b \neq \emptyset$, where $a, b \in A(H)$ include x, y respectively. Thus K may be chosen so that $x, y \in ins(W)$. Now no atom of H which intersects $ins(W)$ also intersects Z, since $d(x, y) < d(x, \sigma)$ for all $\sigma \in Z$. Consequently every atom of H which intersects $ins(W)$ is an atom of H', and we may therefore choose a radial drawing K' of H' such that W is a walk of K'. If $d(x, y) \geq ord(\mathcal{T}')$ then certainly $d(x, y) \geq d'(x, y)$ as required; and so we may assume that $d(x, y) < ord(\mathcal{T}')$, and $ins'(W)$ is defined. By (4.1), $ins(W) = ins'(W)$, and so $x, y \in ins'(W)$, and $d'(x, y) \leq d(x, y)$ as required. □

Let H be a 2-cell drawing in Σ, let $\mathcal{T}$ be a respectful tangle in H of order θ with metric d, let $\lambda \geq 0$ be an integer with $\lambda < \theta$, let $z \in A(H)$, and let

H' be the subdrawing of H formed by all edges e of H with $d(z,e) > \lambda$, and their ends (or $H' = H$, if $E(H) = \emptyset$). We define $H - z^\lambda = H'$ (we shall only use this notation when the hidden parameters $\Sigma, \mathcal{T}$ are obvious). [4, Theorem (7.11)] implies that

(4.3) *If $H, \Sigma, \mathcal{T}, \theta, d, \lambda, z$ are as above then $H - z^\lambda$ is 2-cell.*

[4, Theorem (7.10)] implies

[p. 126] (4.4) *Let $\mathcal{T}$ be a respectful tangle of order θ in a 2-cell drawing H in a surface Σ. Let $z \in A(H)$, let $\lambda \geq 2$ be an integer with $\theta > 4\lambda + 2$, and let $H' \subseteq H$ be a 2-cell subdrawing such that $a \in A(H')$ for every $a \in A(H)$ with $d(z,a) > \lambda$. Then there is a unique respectful tangle $\mathcal{T}'$ in H' which is a $(4\lambda+2)$-compression of $\mathcal{T}$, and it includes $(A \cap H', B \cap H')$ for every $(A,B) \in \mathcal{T}$ of order $< \theta - 4\lambda - 2$.*

(4.5) *Let $\mathcal{T}$ be a respectful tangle of order θ in a 2-cell drawing H in a surface Σ. Let $z \in A(H)$, let $\lambda \geq 2$ be an integer with $\theta > 4\lambda + 2$, and let H' be a 2-cell drawing in Σ such that $a \in A(H')$ for every $a \in A(H)$ with $d(z,a) > \lambda$. Then*

(i) *there is a unique $(4\lambda+2)$-compression $\mathcal{T}'$ of $\mathcal{T}$ in H'*

(ii) *$H - z^\lambda \subseteq H'$, and $\mathcal{T}'$ is the tangle induced in H' from the unique $(4\lambda+2)$-compression of $\mathcal{T}$ in $H - z^\lambda$*

(iii) *if $H' \subseteq H$ then $(A \cap H', B \cap H') \in \mathcal{T}'$ for every $(A,B) \in \mathcal{T}$ of order $< \theta - 4\lambda - 2$.*

Proof. By (4.3) and (4.4), there is a unique $(4\lambda+2)$-compression of $\mathcal{T}$ in $H - z^\lambda$, $\mathcal{T}_0$ say. Since $d(z,e) > \lambda$ for every edge e of $H - z^\lambda$ and $H - z^\lambda$ has no isolated vertices (unless $E(H) = \emptyset$, when $\lambda = 0$ and $H = H'$), it follows that $H - z^\lambda \subseteq H'$. Let $\mathcal{T}''$ be the tangle in H' induced by $\mathcal{T}_0$. Then $\mathcal{T}''$ is a $(4\lambda+2)$-compression, and hence $\mathcal{T}'' = \mathcal{T}'$, by (4.1). This proves (i) and (ii), and (iii) follows from (4.4). □

(4.6) *Let $\mathcal{T}, \theta, H, \Sigma, z, \lambda, H', \mathcal{T}'$ be as in (4.5). Let $\mathcal{T}^*$ be a tangle in a graph G, and let $H, \eta, \mathcal{T}$ be a Σ-span in G with respect to $\mathcal{T}^*$. Let H' be rigid, and let η' be an isomorphism from H' to a subgraph of G, such that $\eta'(x) = \eta(x)$ for every vertex or edge x of H with $d(z,x) > \lambda$. Then $H', \eta', \mathcal{T}'$ is a Σ-span of order $\theta - 4\lambda - 2$, obtained from $H, \eta, \mathcal{T}$ by rearranging within λ of z.*

Proof. We must verify conditions (i)-(iv) in the definition of Σ-span. Conditions (i) and (ii) are trivial, and since $\mathcal{T}'$ is a $(4\lambda+2)$-compression it is by definition respectful, and so (iii) holds. It remains to show (iv). Let $(A^*, B^*) \in \mathcal{T}^*$ with order $< \theta - 4\lambda - 2$. We must show that

$$(\eta'^{-1}(A^* \cap \eta'(H')), \eta'^{-1}(B^* \cap \eta'(H'))) \in \mathcal{T}'.$$

Let $H'' = H - z^\lambda$, and let $\mathcal{T}''$ be the unique $(4\lambda+2)$-compression of $\mathcal{T}$ in H''. By (4.5)(ii), $\mathcal{T}'$ is the tangle induced in H' by $\mathcal{T}''$; and so it suffices to show that

$$(\eta'^{-1}(A^* \cap \eta'(H')) \cap H'', \eta'^{-1}(B^* \cap \eta'(H')) \cap H'') \in \mathcal{T}''.$$

[p. 127] But

$$\eta'^{-1}(A^* \cap \eta'(H')) \cap H'' = \eta'^{-1}(A^* \cap \eta'(H') \cap \eta'(H'')) = \eta'^{-1}(A^* \cap \eta'(H''))$$

since $H'' \subseteq H'$ by (4.5)(ii); and $\eta'^{-1}(A^* \cap \eta'(H'')) = \eta^{-1}(A^* \cap \eta(H''))$, because the restriction of η and η' to H'' are equal. Similarly,

$$\eta'^{-1}(B^* \cap \eta'(H')) \cap H'' = \eta^{-1}(B^* \cap \eta(H'')),$$

and so it suffices to show that

$$(\eta^{-1}(A^* \cap \eta(H'')), \eta^{-1}(B^* \cap \eta(H''))) \in \mathcal{T}''.$$

Let $A = \eta^{-1}(A^* \cap \eta(H))$, $B = \eta^{-1}(B^* \cap \eta(H))$. Then $(A, B) \in \mathcal{T}$ since $H, \eta, \mathcal{T}$ is a Σ-span. Moreover, (A, B) has order at most that of (A^*, B^*) and hence less than $\theta - 4\lambda - 2$. By (4.5)(iii), $(A \cap H'', B \cap H'') \in \mathcal{T}''$. But

$$\begin{aligned} A \cap H'' &= \eta^{-1}(A^* \cap \eta(H)) \cap H'' = \eta^{-1}(A^* \cap \eta(H) \cap \eta(H'')) \\ &= \eta^{-1}(A^* \cap \eta(H'')) \end{aligned}$$

and similarly $B \cap H'' = \eta^{-1}(B^* \cap \eta(H''))$, and so

$$(\eta^{-1}(A^* \cap \eta(H'')), \eta^{-1}(B^* \cap \eta(H''))) \in \mathcal{T}''$$

as required. □

We shall also need the following.

(4.7) *Let $\mathcal{T}$ be a respectful tangle of order θ in a 2-cell drawing H in a surface Σ. Let $z \in A(H)$, let $\lambda \geq 2$ be an integer, and let $X \subseteq V(H)$ be free with respect to $\mathcal{T}$. Suppose that $d(z, v) > 2|X| + 5\lambda + 2$ for all $v \in X$. Now let H' be a 2-cell drawing in Σ such that $a \in A(H')$ for every $a \in A(H)$ with $d(z, a) > \lambda$; and let $\mathcal{T}'$ be a $(4\lambda+2)$-compression of $\mathcal{T}$ in H'. Then X is free with respect to $\mathcal{T}'$.*

Proof. Suppose not; then there exists $(A, B) \in \mathcal{T}'$ with $|X \cap V(A)| > |V(A \cap B)|$. Choose such (A, B) with A minimal.

(1) *A is connected.*

Subproof. Suppose that $A = A_1 \cup A_2$, where A_1, A_2 are non-null and $A_1 \cap A_2$ is null. Then $(A_1, A_2 \cup B), (A_2, A_1 \cup B) \in \mathcal{T}$, and by the minimality of A,

$$|X \cap V(A_1)| \leq |V(A_1 \cap (A_2 \cup B))| = |V(A_1 \cap B)|$$
$$|X \cap V(A_2)| \leq |V(A_2 \cap (A_1 \cup B))| = |V(A_2 \cap B)|.$$

[p. 128] Adding, we deduce that $|X \cap V(A)| \leq |V(A \cap B)|$, a contradiction. This proves (1).

Now $|X \cap V(A)| > |V(A \cap B)| \geq 0$, and so we may choose $v \in X \cap V(A)$.

(2) *For every $u \in V(A)$, $d'(u, v) \leq 2|X| - 2$.*

Subproof. By (1) there is a path P of A from u to v. By [4, Theorem (4.1)], $|V(A \cap B)| \geq (1/2)d'(u, v)$, and so $d'(u, v) \leq 2|V(A \cap B)| \leq 2|X| - 2$. This proves (2).

(3) *Every $u \in V(A)$ belongs to $V(H)$ and every edge of H' incident with u is an edge of H.*

Subproof. Let $e \in E(H')$ be incident with $u \in V(A)$. Let $\sigma \in \Sigma$ with $\sigma \in e$. By (2), $d'(\sigma, v) \leq 2|X|$. Hence $d(\sigma, v) \leq 2|X| + 4\lambda + 2$, since $\mathscr{T}'$ is a $(4\lambda + 2)$-compression of $\mathscr{T}$. Since $d(z, v) > 2|X| + 5\lambda + 2$, it follows that $d(z, \sigma) > \lambda$. Let a be the atom of H with $\sigma \in a$. Then $d(z, a) > \lambda$, and so $a \in A(H')$. Hence $a = e$, and so $e \in E(H)$. Similarly, by setting $\sigma = u$ and applying (2), we deduce that $u \in V(H)$. This proves (3).

From (3), A is a subgraph of H.

(4) *For all $u \in V(A)$, every edge of H incident with u is an edge of H'.*

Subproof. Let $e \in E(H)$ be incident with u. Since $d'(u, v) \leq 2|X| - 2$ and hence $d(u, v) \leq 2|X| - 2 + 4\lambda + 2$, it follows that $d(z, u) > \lambda$ since $d(z, v) > 2|X| + 5\lambda$. Hence $d(z, e) > \lambda$, and so $e \in E(H')$. This proves (4).

By (3) and (4), there is a subgraph C of H such that (A, C) is a separation of H and $A \cap B = A \cap C$. Now $H_0 = H - z^\lambda$ is a subgraph of both H and H'; let $\mathscr{T}_0$ be the $(4\lambda + 2)$-compression of $\mathscr{T}$ in H_0. Since $\theta > 4\lambda + 2$, it follows from (4.4) that $(A \cap H_0, B \cap H_0) \in \mathscr{T}_0$, since (A, B) has order $< |X| \leq \theta - 4\lambda - 2$. Since by (4.5)(ii) $\mathscr{T}'$ is the tangle in H' induced by $\mathscr{T}_0$, and $C \cap H_0 = B \cap H_0$, it follows that $(A, C) \in \mathscr{T}'$. But

$$|X \cap V(A)| > |V(A \cap B)| = |V(A \cap C)|$$

and so X is not free with respect to $\mathscr{T}$, by [2, Theorem (12.2)], a contradiction. This completes the proof. □

5. Flatness

In this section we prove that, if λ is large enough, rearranging a (λ, μ)-flat Σ-span produces another Σ-span which is still reasonably flat. First, we need the following.

[p. 129] (5.1) *Let $\mathscr{T}^*$ be a tangle in G, and let $H, \eta, \mathscr{T}$ be a (λ, μ)-flat Σ-span of order θ. Let $0 \le \lambda_1 \le \lambda$, let $\mu_1 \ge \mu$ and let $1 \le \theta_1 \le \theta$, such that $\theta_1 \ge 4\lambda_1 + \mu_1 + 2$; and let $\mathscr{T}_1$ be the θ_1-truncation of $\mathscr{T}$. Then $(H, \eta, \mathscr{T}_1)$ is (λ_1, μ_1)-flat.*

Proof. Certainly $(H, \eta, \mathscr{T}_1)$ is a Σ-span of order θ_1, and $\theta_1 \ge 4\lambda_1 + \mu_1 + 2$. Suppose it is not (λ_1, μ_1)-flat. Let $H', \eta', \mathscr{T}_1'$ be obtained from $H, \eta, \mathscr{T}_1$ by rearranging within λ_1 of some $z \in A(H)$, and let P be an $\eta'(H')$-path in G with ends $\eta'(u), \eta'(v)$, where $d'(u, v) \ge \mu_1$.

(1) *$a \in A(H')$ for every $a \in A(H)$ with $d(z, a) > \lambda$.*

Subproof. Now $d_1(z, a) = \min(\theta_1, d(z, a))$, and since $\theta_1 > \lambda_1$ and $d(z, a) > \lambda \ge \lambda_1$, it follows that $d_1(z, a) > \lambda_1$. Hence $a \in A(H')$, since $H', \eta', \mathscr{T}_1'$ is obtained from $H, \eta, \mathscr{T}_1$ by rearranging within λ_1 of z. This proves (1).

Since $\mathscr{T}_1'$ is a $(4\lambda_1+2)$-compression of $\mathscr{T}_1$, and $\mathscr{T}_1$ is a $(\theta-\theta_1)$-compression of $\mathscr{T}$, it follows that $\mathscr{T}_1'$ is a $(\theta - \theta_1 + 4\lambda_1 + 2)$-compression of $\mathscr{T}$. Let $\mathscr{T}'$ be the $(4\lambda + 2)$-compression of $\mathscr{T}$ in H' (this exists by (4.5)). By (4.1), if $\theta - 4\lambda - 2 \ge \theta_1 - 4\lambda_1 - 2$ then $\mathscr{T}_1' \subseteq \mathscr{T}'$, and if $\theta - 4\lambda - 2 \le \theta_1 - 4\lambda_1 - 2$ then $\mathscr{T}' \subseteq \mathscr{T}_1'$.

(2) *$d'(u, v) \ge \mu$.*

Subproof. Certainly $d_1'(u, v) \ge \mu_1 \ge \mu$. If $\mathscr{T}' \subseteq \mathscr{T}_1'$, then

$$d'(u, v) = \min(\theta - 4\lambda - 2, d_1'(u, v)) \ge \mu$$

since $\theta \ge 4\lambda + \mu$ (because $H, \eta, \mathscr{T}$ is (λ, μ)-flat). If $\mathscr{T}_1' \subseteq \mathscr{T}'$ then $d'(u, v) \ge d_1'(u, v) \ge \mu$. This proves (2).

From (1) and (2), $H', \eta', \mathscr{T}'$ is a μ-stepped Σ-span, obtained from $H, \eta, \mathscr{T}$ by rearranging within λ of z, which is impossible since $H, \eta, \mathscr{T}$ is (λ, μ)-flat. The result follows. □

The main result of this section is the following.

(5.2) *Let $\mathscr{T}^*$ be a tangle in a graph G, let Σ be a surface, and let $H, \eta, \mathscr{T}$ be a (λ, μ)-flat Σ-span of order θ. Let $\lambda_1, \lambda_2 \ge 4$ and $\mu \ge \emptyset 1$ be integers[F12] such that $\lambda \ge 6\lambda_1 + 5\lambda_2 + 3\mu + 4$. Let $z_1 \in A(H)$, and let $H', \eta', \mathscr{T}'$ be a Σ-span obtained from $H, \eta, \mathscr{T}$ by rearranging within λ_1 of z_1. Then $H', \eta', \mathscr{T}'$ is (λ_2, μ)-flat.*

[F12] **Degenerate case.** As printed, (5.2) fails for $\mu = 0$. Every pair of vertices has $d(u, v) \ge 0$, so every Σ-span with an $\eta'(H')$-path is 0-stepped, and hence not $(\lambda_2, 0)$-flat. Also, for $\theta = 4\lambda + 2$, the truncation $\mathscr{T}^0$ in the proof of Claim (3) would have order 0. For $\mu \ge 1$, (5.2) holds, and all its uses have $\mu \ge 12$.

Proof. Let $\mathscr{T}$ have order θ; then $\mathscr{T}'$ has order $\theta-4\lambda_1-2$. Let $z_2 \in A(H')$, and let $H'', \eta'', \mathscr{T}''$ be obtained from $H', \eta', \mathscr{T}'$ by rearranging within λ_2 of z_2; and suppose, for a contradiction, that $H'', \eta'', \mathscr{T}''$ is μ-stepped. Let P be an $\eta''(H'')$-path in G with ends $\eta''(u), \eta''(v)$, where [p. 130] $u, v \in V(H'')$ and $d''(u,v) \geq \mu$. Since $\mathscr{T}'$ is a $(4\lambda_1+2)$-compression of $\mathscr{T}$, and $\mathscr{T}''$ is a $(4\lambda_2+2)$-compression of $\mathscr{T}'$, we deduce

(1) *For all* $\sigma_1, \sigma_2 \in \Sigma$, $d'(\sigma_1,\sigma_2) \geq d(\sigma_1,\sigma_2)-4\lambda_1-2$ *and* $d''(\sigma_1,\sigma_2) \geq d'(\sigma_1,\sigma_2) - 4\lambda_2 - 2$.

Let Z_1 be the union of all atoms a of H with $d(z_1,a) \leq \lambda_1$, and let Z_2 be the union of all atoms a of H' with $d'(z_1,a) \leq \lambda_2$. Then, again immediately from the definitions, we have

(2) Z_1 *includes every* $a \in A(H) - A(H')$, *and* Z_2 *includes every* $a \in A(H') - A(H'')$.

Choose $\tau \in \Sigma$ with $\tau \in z_2$, and let $z_3 \in A(H)$ with $\tau \in z_3$.

(3) $d(z_1,z_3) > \lambda - \lambda_1$.

Subproof. Suppose not. Let $\mathscr{T}^0$ be the $(\theta-4\lambda-2)$-truncation of $\mathscr{T}''$. (This exists, since $\mathrm{ord}(\mathscr{T}'') = \theta-4(\lambda_1+\lambda_2)-4 \geq \theta-4\lambda-2 \geq 1$.) We claim that $H'', \eta'', \mathscr{T}^0$ is a Σ-span obtained from $H, \eta, \mathscr{T}$ by rearranging within λ of z_3, and it is μ-stepped. Certainly it is a Σ-span, because $H'', \eta'', \mathscr{T}''$ is a Σ-span and $\mathscr{T}^0 \subseteq \mathscr{T}''$; and it is μ-stepped, because

$$d^0(u,v) = \max(\theta-4\lambda-2, d''(u,v)) \geq \mu$$

since $\theta \geq 4\lambda+\mu+2$ and $d''(u,v) \geq \mu$. To see that $H'', \eta'', \mathscr{T}^0$ is obtained from $H, \eta, \mathscr{T}$ by rearranging within λ of z_3, we must check that

(i) $a \in A(H'')$ for all $a \in A(H)$ with $d(z_3,a) > \lambda$

(ii) $\eta(x) = \eta''(x)$ for every vertex or edge x of H with $d(z_3,x) > \lambda$, and

(iii) $\mathscr{T}^0$ is a λ-compression of $\mathscr{T}$.

To show (i), let $a \in A(H)$ with $d(z_3,a) > \lambda$. Since by hypothesis $d(z_1,z_3) \leq \lambda - \lambda_1$, it follows that $d(z_1,a) > \lambda_1$, and so $a \cap Z_1 = \emptyset$. Consequently, by (2), $a \in A(H')$. Now $d(\tau,a) > \lambda$, and so $d'(\tau,a) > \lambda - 4\lambda_1 - 2$ since $\mathscr{T}'$ is a $(4\lambda_1+2)$-compression of $\mathscr{T}$. Since $\lambda - 4\lambda_1 - 2 \geq \lambda_2$, it follows that $a \cap Z_2 = \emptyset$ and so $a \in A(H'')$, by (2). This proves (i), and (ii) follows immediately. For (iii), let $\sigma_1, \sigma_2 \in \Sigma$. Then

$$\begin{aligned} d^0(\sigma_1,\sigma_2) &= \min(\theta-4\lambda-2, d''(\sigma_1,\sigma_2)) \\ &\geq \min(\theta-4\lambda-2, d(\sigma_1,\sigma_2) - 4(\lambda_1+\lambda_2) - 4) \end{aligned}$$

by (1); and

$$\min(\theta-4\lambda-2, d(\sigma_1,\sigma_2) - 4(\lambda_1+\lambda_2) - 4) \geq d(\sigma_1,\sigma_2) - 4\lambda - 2$$

[p. 131] because $\theta \geq d(\sigma_1, \sigma_2)$ and $\lambda_1 + \lambda_2 < \lambda$. This proves (iii), and hence proves that $H'', \eta'', \mathscr{T}^0$ is obtained from $H, \eta, \mathscr{T}$ by rearranging within λ of z_3. But it is μ-stepped, a contradiction since $H, \eta, \mathscr{T}$ is (λ, μ)-flat. This proves (3).

(4) *For all* $\sigma_1 \in Z_1$, $d(\tau, \sigma_1) > \lambda - 2\lambda_1$, *and hence* $d'(\tau, \sigma_1) > \lambda - 6\lambda_1 - 2$.

Subproof. By (3), $\lambda - \lambda_1 < d(z_1, \tau) \leq d(z_1, \sigma_1) + d(\sigma_1, \tau) \leq \lambda_1 + d(\sigma_1, \tau)$, and so the first inequality holds. The second follows since $\mathscr{T}'$ is a $(4\lambda_1 + 2)$-compression of $\mathscr{T}$. This proves (4).

(5) *For all* $\sigma_2 \in Z_2$, $d(\tau, \sigma_2) \leq \lambda_2$.

Subproof. For all $\sigma_1 \in Z_1$, $d'(\tau, \sigma_1) > \lambda - 6\lambda_1 - 2 \geq \lambda_2 \geq d'(\tau, \sigma_2)$ by (4). Hence, by (4.2) (applied to $\mathscr{T}, \mathscr{T}', Z_1$), $d(\tau, \sigma_2) \leq d'(\tau, \sigma_2) \leq \lambda_2$. This proves (5).

(6) *For all* $\sigma_1 \in Z_1$ *and* $\sigma_2 \in Z_2$, $d(\sigma_1, \sigma_2) > \lambda - 2\lambda_1 - \lambda_2$.

Subproof. Since $d(z_1, \tau) \leq d(z_1, \sigma_1) + d(\sigma_1, \sigma_2) + d(\sigma_2, \tau)$, the claim follows from (3) and (5).

In particular, from (6) $Z_1 \cap Z_2 = \emptyset$ and so $z_2 = z_3$. Let $H_0 \subseteq H$ be the subdrawing formed by the edges of H not in $Z_1 \cup Z_2$ and their ends. Let L_1, M_1, L_2, M_2 be the subdrawings of H, H', H, H'' formed by its edges in Z_1, Z_1, Z_2, Z_2 and their ends, respectively. Then $H = H_0 \cup L_1 \cup L_2$, $H' = H_0 \cup M_1 \cup L_2$, $H'' = H_0 \cup M_1 \cup M_2$.

(7) L_1 *and* L_2 *both have circuits.*

Subproof. There is a region r of H with $d(z_1, r) \leq 2$; and hence $d(z_1, e) \leq 4 \leq \lambda_1$ and so $e \subseteq \not{\Lambda}_1 Z_1$,[F13] for every edge e of H incident with r. Since H is not a forest (because it has a tangle of order ≥ 3) this set of edges includes a circuit. Similarly L_2 has a circuit, since $\lambda_2 \geq 4$. This proves (7).

Let $H''' = H_0 \cup L_1 \cup M_2$.

(8) H''' *is rigid and 2-cell.*

Subproof. Let $F \subseteq \Sigma$ be an O-arc with $F \cap U(H''') \subseteq V(H''')$ and $|F \cap V(H''')| \leq 2$. Suppose first that $F \cap \not{\Lambda}_2 Z_2 = \emptyset$.[F14] Then $F \cap U(H) \subseteq V(H)$ and $|F \cap V(H)| \leq 2$, and so there is a dial $\Delta \subseteq \Sigma$ for F, H. Now by (7), $Z_2 \not\subseteq \Delta$, because L_2 has a circuit, and so $Z_2 \cap \Delta = \emptyset$, and Δ is a dial for F, H''' as required. Thus we may assume that $F \cap Z_2 \neq \emptyset$. Suppose now that $F \cap Z_1 = \emptyset$; then a similar argument applied to H'' yields the desired dial. Finally, we assume that $F \cap Z_2 \neq \emptyset$ and $F \cap Z_1 \neq \emptyset$. But $|F \cap U(H''')| \leq 2$, and this is impossible by (6). This proves that H''' is rigid.

[F13] **Misprint.** Here and in Claim (8), the sets Z_1 and Z_2 defined in the proof are meant, as the next sentences show.

[F14] **Misprint.** The set Z_2 is meant, as in the note to Claim (7).

[p. 132] Since $bd(\Sigma) = \emptyset$ and H''' is non-null, it follows that H''' is 2-cell. This proves (8).

(9) *Every component of $\eta'(M_1)$ meets $\eta(L_1)$, and every component of $\eta''(M_2)$ meets $\eta'(L_2) = \eta(L_2)$.*

Subproof. Let C be a component of M_1 (so $\eta'(C)$ is a component of $\eta'(M_1)$). Now $M_1 \neq H'$ since L_2 is not null by (7), and since H' is 2-cell and hence connected and $Z_1 \cap Z_2 = \emptyset$, there is a vertex $x \in V(M_1)$ incident with an edge e of H' not in $E(M_1)$, and with its other end not in $V(M_1)$. Then $e \in E(H)$, and so $x \in V(H)$, and since $x \in Z_1$ it follows that $x \in V(L_1)$. Moreover, $\eta(x) = \eta'(x)$ (because $\eta(e) = \eta'(e)$ and $\eta(y) = \eta'(y)$ where e has ends x, y). Hence $\eta(x) \in V(\eta'(C)) \cap V(\eta(L_1))$, and so $\eta'(C)$ meets $\eta(L_1)$, as required. The argument for $\eta''(M_2)$ is similar. This proves (9).

Let X_1 be the union of $V(\eta(L_1)) \cup V(\eta'(M_1))$ with the set

$$\{\eta(x) : x \in V(H) \text{ and } d(\sigma_1, x) < \mu \text{ for some } \sigma_1 \in Z_1\}.$$

Let X_2 be the union of $V(\eta(L_2)) \cup V(\eta''(M_2))$ with

$$\{\eta(x) : x \in V(H) \text{ and } d(\sigma_2, x) < \mu \text{ for some } \sigma_2 \in Z_2\}.$$

Then $X_1, X_2 \subseteq V(G)$.

(10) *Every path of G from X_1 to X_2 has an internal vertex or edge in $\eta(H)$.*

Subproof. Suppose that some path of G from X_1 to X_2 has no internal vertex or edge in $\eta(H)$. By (9), there is an $\eta(H)$-path with ends $\eta(t_1), \eta(t_2)$ where $d(\sigma_i, t_i) \leq \mu$ for some $\sigma_i \in Z_i$ $(i = 1, 2)$. Since $H, \eta, \mathcal{T}$ is $(0, \mu)$-flat by (5.1), it follows that $d(t_1, t_2) < \mu$, and so $d(\sigma_1, \sigma_2) < 3\mu \leq \lambda - 2\lambda_1 - \lambda_2$ contrary to (6). This proves (10).

It follows from (10) that $\eta(L_1)$ is disjoint from $\eta''(M_2)$. For each vertex or edge x of H_1, define $\eta'''(x) = \eta(x)$ unless x belongs to M_2, when $\eta'''(x) = \eta''(x)$. Then η''' is an isomorphism from H''' to the subgraph $\eta(H_0) \cup \eta(L_1) \cup \eta''(M_2)$ of G.

(11) *$a \in A(H''')$ for every $a \in A(H)$ with $d(z_2, a) > \lambda_2$, and for every vertex or edge x of H''', if $d(z_2, x) > \lambda_2$ then $\eta'''(x) = \eta(x)$.*

Subproof. If $a \in A(H)$ with $d(z_2, a) > \lambda_2$, then $a \cap Z_2 = \emptyset$ by (5), and so $a \in A(H''')$. The second claim follows similarly.

Let $\mathcal{T}'''$ be the (unique, by (4.5)) $(4\lambda_2 + 2)$-compression of $\mathcal{T}$ in H'''. Then from (4.6), (8) and (11), we have

[p. 133] (12) *$H''', \eta''', \mathcal{T}'''$ is a Σ-span of order $\theta - 4\lambda_2 - 2$ obtained from $H, \eta, \mathcal{T}$ by rearranging within λ_2 of z_2.*

We recall that P is an $\eta''(H'')$-path with ends $\eta''(u), \eta''(v)$, where $d''(u,v) \geq \mu$.

(13) *Either P meets $\eta(L_2) \cup \eta''(M_2)$, or $d(u, \sigma_2) < \mu$ for some $\sigma_2 \in Z_2$.*

Subproof. Suppose that P does not meet $\eta(L_2) \cup \eta''(M_2)$. Then $u, v \in V(M_1 \cup H_0)$, and P is an $\eta'(H')$-path. Since $H, \eta, \mathcal{T}$ is (λ, μ)-flat, and $H', \eta', \mathcal{T}'$ is obtained from $H, \eta, \mathcal{T}$ by rearranging within $\lambda_1 \leq \lambda$ of z, it follows that $d'(u,v) < \mu$. Since $d''(u,v) \geq \mu$, we deduce from (4.2) (applied to $\mathcal{T}'$, $\mathcal{T}''$ and Z_2) that $d'(u,v) \geq d'(u, \sigma_2)$ for some $\sigma_2 \in Z_2$, and consequently $d'(u, \sigma_2) < \mu$. By (4), for all $\sigma_1 \in Z_1$,

$$d'(\sigma_1, \sigma_2) \geq d'(\tau, \sigma_1) - d'(\tau, \sigma_2) > \lambda - 6\lambda_1 - \lambda_2 - 2 \geq \mu > d'(u, \sigma_2)$$

and so by (4.2) (applied to $\mathcal{T}$, $\mathcal{T}'$ and Z_1), $d(u, \sigma_2) \leq d'(u, \sigma_2) < \mu$. This proves (13).

(14) *Either P meets $\eta(L_1) \cup \eta'(M_1)$, or $d(u, \sigma_1) < \mu$ for some $\sigma_1 \in Z_1$.*

Subproof. Suppose that P does not meet $\eta(L_1) \cup \eta'(M_1)$. Then P is an $\eta'''(H''')$-path, and so $d'''(u,v) < \mu$ by (12), since $\lambda_2 \leq \lambda$. Let W be a closed walk of a radial drawing of H''' with length $< 2\mu$ and with $u, v \in ins'''(W)$. Suppose first that $Z_1 \cap ins'''(W) = \emptyset$. Then W is a walk of a radial drawing of H'', and so $ins''(W) = ins'''(W)$ by (4.1), since $\mathcal{T}''$ and $\mathcal{T}'''$ are $(4\lambda_1 + 4\lambda_2 + 4)$- and $(4\lambda_2 + 2)$-compressions of $\mathcal{T}$ respectively. Hence $u, v \in ins''(W)$, and so $d''(u,v) \leq d'''(u,v) < \mu$, a contradiction. We deduce that $Z_1 \cap ins'''(W) \neq \emptyset$, and so there exists $\sigma_1 \in Z_1$ with $d'''(u, \sigma_1) < \mu$. Now Z_2 includes all atoms of H which are not atoms of H'''. Moreover, for all $\sigma_2 \in Z_2$, since $\mathcal{T}'''$ is a $(4\lambda_2 + 2)$-compression of $\mathcal{T}$, and by (6),

$$d'''(\sigma_1, \sigma_2) \geq d(\sigma_1, \sigma_2) - 4\lambda_2 - 2 > (\lambda - 2\lambda_1 - \lambda_2) - 4\lambda_2 - 2 \geq \mu > d'''(\sigma_1, u).$$

Hence by (4.2) applied to $\mathcal{T}$, $\mathcal{T}'''$ and Z_2, we deduce that $d(\sigma_1, u) \leq d'''(\sigma_1, u) < \mu$. This proves (14).

From (13) and (14) it follows that P meets X_1 and X_2; but this contradicts (10). We deduce that there is no such P, that is, $H'', \eta'', \mathcal{T}''$ is not μ-stepped. Since $\mathcal{T}'$ has order $\theta - 4\lambda_1 - 2 \geq 4\lambda_2 + \mu + 2$, it follows that $H', \eta', \mathcal{T}'$ is (λ_2, μ)-flat, as required. □

6. Crooked Transactions

If $\mathcal{P}$ is a transaction in a society (G, Ω), we denote the set of ends of members of $\mathcal{P}$ by $V(\mathcal{P})$. We say $\mathcal{P}$ is crooked if for every $P \in \mathcal{P}$ with ends [p. 134] u, v say, there exists $u', v' \in V(\mathcal{P})$ so that u, u', v, v' are distinct and occur in that order in Ω.

Let α be an arrangement in a surface Σ of a segregation $\mathcal{S}$ of G. A subset $X \subseteq \Sigma$ is said to be α-normal if

$$X \cap \alpha(A, \Omega) \subseteq \{\alpha(v) : v \in \bar{\Omega}\}$$

for all $(A, \Omega) \in \mathcal{S}$. Let $F \subseteq \Sigma$ be an α-normal O-arc. Let $X = \{v \in V(\mathcal{S}) : \alpha(v) \in F\}$ and $X' = \{\alpha(v) : v \in X\}$. Let Ω' be a natural order of X' from F, and let Ω be the cyclic permutation with $\bar{\Omega} = X$ mapped to Ω' by α. We define $\alpha^{-1}(F)$ to be Ω (or its reverse; the choice will not matter). We shall need the following theorem [1, Theorem (11.11)].

(6.1) *Let $\pi \geq 2$ be an integer, with $\pi \neq 3$, and let (G, Ω) be a society with no crooked transaction of cardinality π. Then there is a segregation $\mathcal{S}$ of G of type $(3\pi + 9, 1)$ with $\bar{\Omega} \subseteq V(\mathcal{S})$, and an arrangement α of $\mathcal{S}$ in a closed disc Δ, such that $\Omega = \alpha^{-1}(bd(\Delta))$.*

Let $\mathcal{P} = \{P_1, \ldots, P_\pi\}$ be a transaction in a society (G, Ω), where P_i has ends a_i, b_i $(1 \leq i \leq \pi)$. If

(i) $\pi \geq 2$ *and* $a_\pi, a_{\pi-1}, \ldots, a_1, b_\pi, b_{\pi-1}, \ldots, b_1$ *occur in* Ω *in that order, $\mathcal{P}$ is a crosscap transaction*

(ii) $\pi \geq 3$ *and* $a_\pi, a_{\pi-1}, \ldots, a_2, a_1, b_2, b_3, \ldots, b_\pi, b_1$ *occur in* Ω *in that order, $\mathcal{P}$ is a leap transaction*

(iii) $\pi \geq 4$ *and* $a_\pi, a_{\pi-1}, \ldots, a_2, a_1, b_2, b_1, b_3, b_4, \ldots, b_{\pi-3}, b_{\pi-2}, b_\pi, b_{\pi-1}$ *occur in* Ω *in that order, $\mathcal{P}$ is a doublecross transaction.*

We shall need the following [1, Theorem (7.1)].

(6.2) *Let $\phi \geq 2$, $\tau_1 \geq 3$, $\tau_2 \geq 4$ be integers. Let $\mathcal{P}$ be a crooked transaction in a society (G, Ω) such that $|\mathcal{P}| \geq (\phi-1)(2\tau_1+\tau_2-7)$.*[F15] *Then there exists $\mathcal{P}' \subseteq \mathcal{P}$ such that either*

(i) $|\mathcal{P}'| = \phi$ *and $\mathcal{P}'$ is a crosscap transaction, or*

(ii) $|\mathcal{P}'| = \tau_1$ *and $\mathcal{P}'$ is a leap transaction, or*

(iii) $|\mathcal{P}'| = \tau_2$ *and $\mathcal{P}'$ is a doublecross transaction.*

7. The Main Proof

Now we shall prove (1.2). Throughout this section, let Σ be a surface with $bd(\Sigma) = \emptyset$, and let $k \geq 0$, $\phi \geq 8$, $\mu \geq 12$, $\lambda_{k+1} \geq 4$ and $\theta_{k+1} \geq 4\lambda_{k+1} + \mu + 2$ be fixed integers, with μ even. We shall show that they satisfy

[F15] **Misprint.** No i is bound in the statement. Section 7, which applies (6.2), sets $\pi = (\phi - 1)(2\tau_1 + \tau_2 - 7)$.

[p. 135] (1.2). To do so, we may assume that θ_{k+1} is even and $\theta_{k+1} > 6\mu+14$, by increasing θ_{k+1} if necessary. Let $\pi = (\phi-1)(47+\mu+\theta_{k+1}/2)$[F16], let $\beta = 2\pi+8$, and let $\theta_k = 100\beta$, $\lambda_k = 20\beta$, $\rho_k = 3\pi+9$. We claim that $\theta_k, \lambda_k, \rho_k$ satisfy (1.2). (For convenience, let us write $\theta, \theta', \lambda, \lambda', \rho$ for $\theta_k, \theta_{k+1}, \lambda_k, \lambda_{k+1}, \rho_k$ respectively.) To show this, let (again, throughout this section) $\mathcal{T}^*$ be a tangle in a graph G such that there is a (λ,μ)-flat Σ-span of order $\geq \theta$, with k eyes mutually at distance $\geq \theta$. We must prove that one of the following statements (S1), (S2), (S3) holds.

(S1) *There is a (λ',μ)-flat Σ-span of order θ' with $k+1$ eyes, mutually at distance θ'.*

(S2) *There is a Σ'-span of order ϕ, where Σ' is a surface obtained by adding a crosscap to Σ.*

(S3) *There is a $\mathcal{T}^*$-central segregation of G of type (ρ,k) with an arrangement in Σ.*

The proof proceeds in a series of lemmas. We begin with the following.

(7.1) *There is a (λ,μ)-flat Σ-span of order θ, with k eyes mutually at distance θ.*

Proof. Let $H, \mathcal{T}$ be a (λ,μ)-flat Σ-span of order $\geq\theta$ with k eyes $r_1,\dots,r_k$ satisfying $d(r_i,r_j)\geq\theta$ $(1\leq i<j\leq k)$. Let $\mathcal{T}'$ be the θ-truncation of $\mathcal{T}$. Then $H,\mathcal{T}'$ is a Σ-span, and it is (λ,μ)-flat by (5.1), since $\theta\geq 4\lambda+\mu+2$. Moreover, $r_1,\dots,r_k$ are eyes of $H,\mathcal{T}'$ since $\mathrm{ord}(\mathcal{T}')=\theta\geq 4$; and for $1\leq i<j\leq k$, $d'(r_i,r_j)=\min(\theta,d(r_i,r_j))=\theta$. The result follows. □

Throughout this section, let $H,\mathcal{T}$ be a (λ,μ)-flat Σ-span of order θ, with metric d, and let $r_1,\dots,r_k$ be eyes of $H,\mathcal{T}$ with $d(r_i,r_j)=\theta$ for $1\leq i<j\leq k$. For $1\leq i\leq k$, let Z_i be the union of all atoms a of H with $d(r_i,a)\leq\theta'+12\mu+14$, and let J_i be the union of all bridges B of H in G with $V(B)\cap Z_i\neq\emptyset$. Let J_0 be the union of all bridges B of H in G such that $V(B)\cap(Z_1\cup\cdots\cup Z_4)=\emptyset$.

(7.2) *For $1\leq i<j\leq k$, if $\sigma_1\in Z_i$ and $\sigma_2\in Z_j$ then $d(\sigma_1,\sigma_2)\geq\theta-2(\theta'+12\mu+14)$, and so $Z_i\cap Z_j=\emptyset$ and $J_i\cap J_j\subseteq H$.*

Proof. We have

$$\theta=d(r_i,r_j)\leq d(r_i,\sigma_1)+d(\sigma_1,\sigma_2)+d(r_j,\sigma_2)\leq d(\sigma_1,\sigma_2)+2(\theta'+12\mu+14)$$

[p. 136] and so the inequality follows. Hence $Z_i\cap Z_j=\emptyset$, since $\theta>2(\theta'+12\mu+14)$. Finally, suppose that $J_i\cap J_j\not\subseteq H$. Then there is an H-path with one end, u, in Z_i and the other, v, in Z_j. Hence

$$d(u,v)\geq\theta-2(\theta'+12\mu+14)\geq\mu.$$

[F16] **Proof gap.** The value changes together with τ_2 on page 138, so that $\pi=(\phi-1)(2\tau_1+\tau_2-7)$ still holds (see the note there).

But $H, \mathcal{T}$ is $(0, \mu)$-flat by (5.1), a contradiction. Thus $J_i \cap J_j \subseteq H$. $\square$

Let $G_0 = H \cup J_0$.

(7.3) *If there is a $(3\mu+3)$-zone $\Lambda \subseteq \Sigma$ for H such that $H \cap (\Sigma - \Lambda)$ is rigid and such that Λ is an eye of $H \cap (\Sigma - \Lambda)$ in G_0 with respect to the tangle of order $\theta - 12\mu - 14$ obtained from $\mathcal{T}$ by clearing Λ, then (S1) holds.*

Proof. Choose $z \in A(H)$ such that Λ is a $(3\mu+3)$-zone around z. Let $\mathcal{T}_1$ be the tangle of order $\theta - 12\mu - 14$ obtained from $\mathcal{T}$ by clearing Λ; then Λ is an eye of $H \cap (\Sigma - \Lambda)$ with respect to $\mathcal{T}_1$. Let $H' = H \cap (\Sigma - \Lambda)$. From the definition of clearing Λ, we have

(1) *For all $\sigma_1, \sigma_2 \in \Sigma$, $d_1(\sigma_1, \sigma_2) \geq d(\sigma_1, \sigma_2) - 12\mu - 14$, and so $\mathcal{T}_1$ is a $(12\mu+14)$-compression of $\mathcal{T}$.*

(2) *For $1 \leq i \leq k$, $\bar{\Lambda} - Z_i$ contains a vertex of H.*

Subproof. Since Λ is an eye of $H \cap (\Sigma - \Lambda)$ in G_0 it follows that some vertex of $V(H) \cap \bar{\Lambda}$ belongs to $V(J_0)$. But no such vertex belongs to Z_i. This proves (2).

(3) *For $1 \leq i \leq k$, $d(z, r_i) > \theta' + 9\mu + 11$.*

Subproof. From (2), there exists $v \in V(H)$ with $v \in \bar{\Lambda} - Z_i$. Then $d(z, v) \leq 3\mu + 3$ since Λ is a $(3\mu+3)$-zone; but $d(r_i, v) > \theta' + 12\mu + 14$, since $v \notin Z_i$. Since $d(r_i, v) \leq d(r_i, z) + d(z, v)$, it follows that $d(r_i, z) > \theta' + \mu + 11$. This proves (3).

In particular, from (3) it follows that $r_i \notin \Lambda$, and so r_i is a region of H', for $1 \leq i \leq k$. Also, Λ is a region of H'.

(4) *For $1 \leq i \leq k$, $d_1(\Lambda, r_i) \geq \theta'$.*

Subproof. Again from (2), let $v \in V(H)$ with $v \in \bar{\Lambda} - Z_i$. Choose $\sigma \in \Lambda$ with $d(v, \sigma) \leq 1$. Then $d(r_i, \sigma) \geq \theta' + 12\mu + 14$, since $d(r_i, v) > \theta' + 12\mu + 14$; and so, by (1), $d_1(r_i, \sigma) \geq \theta'$. This proves (4).

(5) *For $1 \leq i \leq k$, r_i is an eye of the Σ-span $H', \mathcal{T}_1$.*

Subproof. It suffices to show that if $a, b, c, d \in V(H) \cap \mathrm{bd}(r_i)$ are distinct and $\{a, b, c, d\}$ is free with respect to $\mathcal{T}$ then $a, b, c, d \in V(H')$ and [p. 137] $\{a, b, c, d\}$ is free with respect to $\mathcal{T}_1$. The first claim follows from (3). For the second, we observe that, for all $v \in \{a, b, c, d\}$, by (3),

$$d(v, z) \geq d(r_i, z) - 1 \geq \theta' + 9\mu + 11 > 8 + 5(3\mu + 3) + 2.$$

Hence, by (4.7) applied to $\mathcal{T}$ and $\mathcal{T}_1$ (taking $\lambda = 3\mu + 3$) and (1), we deduce that $\{a, b, c, d\}$ is free with respect to $\mathcal{T}_1$. This proves (5).

From (4.6), $H', \mathcal{T}_1$ is a Σ-span of order $\theta - 12\mu - 14$, obtained from $H, \mathcal{T}$ by rearranging within $3\mu + 3$ of z.

(6) *The Σ-span $H', \mathcal{T}_1$ is (λ', μ)-flat.*

Subproof. From (5.2), since $H, \mathcal{T}$ is (λ, μ)-flat, it suffices to check that $3\mu + 3 \geq 4$, $\lambda' \geq 4$, $\lambda \geq 6(3\mu + 3) + 5\lambda' + 3\mu + 4$, and $\theta \geq 4\lambda + \mu + 2$. All these inequalities hold, and so (6) follows.

Let $\mathcal{T}_2$ be the θ'-truncation of $\mathcal{T}_1$. Then $H', \mathcal{T}_2$ is a Σ-span, and by (5.1) and (6), it is (λ', μ)-flat, since $\theta' \geq 4\lambda' + \mu + 2$.

(7) *$r_1, \ldots, r_k, r_{k+1}$ are eyes of $H', \mathcal{T}_2$ in G, and for $1 \leq i < j \leq k+1$, $d_2(r_i, r_j) = \theta'$.*

Subproof. Since $\theta' \geq 4$ and $\mathcal{T}_2$ is the θ'-truncation of $\mathcal{T}_1$, and since by (5) $r_1, \ldots, r_{k+1}$ are eyes of $H', \mathcal{T}_1$ in G, it follows that $r_1, \ldots, r_{k+1}$ are eyes of $H', \mathcal{T}_2$ in G. Now for $1 \leq i < j \leq k+1$,

$$d_2(r_i, r_j) = \min(\theta', d_1(r_i, r_j)).$$

But if $j = k+1$ then $d_1(r_i, r_j) \geq \theta'$ by (4); and if $j \geq k$ then

$$d_1(r_i, r_j) \geq d(r_i, r_j) - 12\mu - 14 = \theta - 12\mu - 14 \geq \theta'$$

by (1). In either case $d_1(r_i, r_j) \geq \theta'$, and so $d_2(r_i, r_j) = \theta'$. This proves (7).

From (7) we see that (S1) holds, as required. □

Let H be a subgraph of G, and let H be a 2-cell drawing in Σ. Let α be an arrangement in Σ of a segregation $\mathcal{S}$ of G. We say that α is *compatible with H if*

(i) $\alpha(v) = v$ for all $v \in V(H) \cap V(\mathcal{S})$, and

(ii) for each $(A, \Omega) \in \mathcal{S}$, $U(H) \cap \mathrm{bd}(\alpha(A, \Omega)) = V(H) \cap \bar{\Omega}$, and $H \cap \alpha(A, \Omega) = H \cap A$.

(7.4) *If (S1) does not hold, there is a $\mathcal{T}$-local 3-segregation $\mathcal{S}_0$ of G_0 and an arrangement α_0 of it in Σ compatible with H.**

[p. 138] *Proof.* Now $\mu \geq 12$ is even and $\theta \geq 16\mu + 17$. Moreover $d(u, v) < \mu$ for every H-path P in G_0 with ends u, v because $H, \mathcal{T}$ is $(0, \mu)$-flat by (5.1). The result follows from (7.3) and [5, Theorem (10.1)] (taking $\beta = \mu$). □

If α is an arrangement in a surface Σ of a segregation $\mathcal{S}$ of G, and $\Delta \subseteq \Sigma$ is a closed disc with $\mathrm{bd}(\Delta)$ α-normal, we define $\alpha^{-1}(\Delta)$ to be the subgraph of G formed by

$$\bigcup (A \colon (A, \Omega) \in \mathcal{S} \text{ and } \alpha(A, \Omega) \subseteq \Delta)$$

together with all $v \in V(\mathcal{S})$ such that $\alpha(v) \in \Delta$.

Henceforth, in this section, we assume that $\mathcal{S}_0$, α_0 are as in (7.4). Let $\tau_1 = 4 + \mu/2$, $\tau_2 = 36 + \theta'/2$[F17]; then $\pi = (\phi - 1)(2\tau_1 + \tau_2 - 7)$. Let $\beta = 2\pi + 8$.

[F17] **Proof gap.** A gap in the proof of (7.11), page 143: only the $\tau - 4$ paths $P_1, \ldots, P_{\tau-4}$

(7.5) *For $1 \le i \le k$ there is a $(3\beta+3)$-zone Λ_i around r_i and a closed disc $\Delta_i \subseteq \Sigma$ with $\mathrm{bd}(\Delta_i)$ α_0-normal, with the following properties:*

(i) *$\Delta_1, \ldots, \Delta_k$ are mutually disjoint, and for $1 \le i \le k$, $V(J_i \cap H) \subseteq \Delta_i$*

(ii) *$d(r_i, x) \le 3\beta + 6$ for all $x \in A(H)$ with $x \cap \Delta_i \neq \emptyset$*

(iii) *$x \subseteq \Delta_i - \mathrm{bd}(\Delta_i)$ for every $x \in A(H)$ with $d(r_i, x) < \beta$*

(iv) *$H \cap \overline{\Sigma - \Delta_i}$ has a circuit, and $H \cap \overline{\Sigma - \Lambda_i}$ is rigid*

(v) *for $1 \le i \le k$, one of the following holds:*

(a) *there is a crooked transaction $\mathcal{P}$ of cardinality π in $(G_0 \cup J_i, \Omega_i)$ (where Ω_i is a natural order of $V(H) \cap \mathrm{bd}(\Lambda_i)$ from $\mathrm{bd}(\Lambda_i)$) such that $V(\mathcal{P})$ is free with respect to the $(12\beta+14)$-compression of $\mathcal{T}$ in $H \cap (\Sigma - \Lambda_i)$, and each $P \in \mathcal{P}$ meets $H \cap (\Sigma - \Lambda_i)$ precisely in the ends of P*

(b) *there is no crooked transaction of cardinality π in $(J_i \cup \alpha_0^{-1}(\Delta_i), \alpha_0^{-1}(\mathrm{bd}(\Delta_i)))$.*

Proof.

(1) *For $1 \le i \le k$, every $v \in V(J_i \cap H)$ satisfies $d(r_i, v) \le \theta' + 13\mu + 13$.*

Subproof. Let B be a bridge of H in G with $B \subseteq J_i$ and $v \in V(B)$. By definition of J_i, there exists $u \in V(B \cap H)$ with $d(r_i, u) \le \theta' + 12\mu + 14$. Now since B is a bridge, there is an H-path in G with ends u, v. Since $H, \mathcal{T}$ is $(0, \mu)$-flat it follows that $d(u, v) < \mu$, and so $d(r_i, v) < \theta' + 13\mu + 14$. This proves (1).

Now $\pi \ge 0$, $\pi \neq 1, 3$, and $\beta = 2\pi + 8$ is even, and $\theta \ge 16\beta + 17$. Since $\theta' + 13\mu + 13 < \beta$, we deduce from [5, Theorem (9.6)] (with the substitutions $z \to r_i$, $G \to G_0 \cup J_i$, $J \to J_i$, $G' \to G_0$, $\mathcal{S} \to \mathcal{S}_0$, $\alpha \to \alpha_0$) that there s a $(3\beta+3)$-zone Λ_i around r_i, and a closed disc $\Delta_i \subseteq \Sigma$ with $\mathrm{bd}(\Delta_i)$ α_0-normal, satisfying (ii)-(v) of the theorem.

[p. 139] From (iii) we see that $V(J_i \cap H) \subseteq \Delta_i$ for $1 \le i \le k$, since $\theta' + 13\mu + 13 < \beta$. Moreover, $\Delta_1, \ldots, \Delta_k$ are mutually disjoint; for if, say $\sigma \in \Delta_i \cap \Delta_j$ where $i \neq j$, then $d(r_i, \sigma), d(r_j, \sigma) \le 3\beta + 6$ by (ii), and so $d(r_i, r_j) \le 6\beta + 12 < \theta$, a contradiction. Hence (i) holds, as required. □

Henceforth we assume that $\Lambda_i, \Delta_i, \Omega_i$ are as in (7.5). For $1 \le i \le k$, let $\Omega_i^* = \alpha_0^{-1}(\mathrm{bd}(\Delta_i))$ and $G_i^* = J_i \cup \alpha_0^{-1}(\Delta_i)$.

(7.6) *$G_1^*, \ldots, G_k^*$ are mutually disjoint.*

Proof. Let $1 \le i < j \le k$, and suppose that $v \in V(G_i^* \cap G_j^*)$. If $v \notin V(\alpha_0^{-1}(\Delta_i))$ then $v \in V(J_i)$ and $v \notin V(H)$, and so $v \notin V(H \cup J_0 \cup J_j)$; and consequently $v \notin V(G_j^*)$, a contradiction. We assume then that $v \in$

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are drawn into H', so (3.4) is applied with $k = \tau - 3$, and it gives $d'(s_1, s_2) \ge 2(\tau-3) - 6 = 2\tau - 12$, not $2\tau - 6$. With $\tau_2 = 3 + \theta'/2$ this is $\theta' - 6 < \theta'$. Taking $\tau_2 = 6 + \theta'/2$, and hence $\pi = (\phi - 1)(7 + \mu + \theta'/2)$, gives $2\tau - 12 = \theta'$. The constants of (1.2) are only claimed to exist, so (1.2) holds as printed.

$V(\alpha_0^{-1}(\Delta_i))$, and similarly $v \in V(\alpha_0^{-1}(\Delta_j))$. Now either $v \in V(A)$ for some $(A,\Omega) \in \mathcal{S}_0$ with $\alpha_0(A,\Omega) \subseteq \Delta_i$, or $v \in V(\mathcal{S}_0)$ and $\alpha_0(v) \in \Delta_i$; and either $v \in V(A')$ for some $(A',\Omega') \in \mathcal{S}_0$ with $\alpha_0(A',\Omega') \subseteq \Delta_j$, or $v \in V(\mathcal{S}_0)$ and $\alpha_0(v) \in \Delta_j$.

Suppose that $v \in V(\mathcal{S}_0)$. Since $\Delta_i \cap \Delta_j = \emptyset$, we may assume that $\alpha_0(v) \notin \Delta_i$. Hence there exists $(A,\Omega) \in \mathcal{S}_0$ with $\alpha_0(A,\Omega) \subseteq \Delta_i$ such that $v \in V(A)$. Since $v \in V(A)$ and $v \in V(\mathcal{S})$, it follows that $v \in \bar{\Omega}$, and hence $\alpha_0(v) \in \mathrm{bd}(\alpha(A,\Omega)) \subseteq \Delta_i$, a contradiction. Hence $v \notin V(\mathcal{S}_0)$.

We deduce that $v \in V(A \cap A')$ for some $(A,\Omega),(A',\Omega') \in \mathcal{S}_0$ with $\alpha_0(A,\Omega) \subseteq \Delta_i$ and $\alpha_0(A',\Omega') \subseteq \Delta_j$. Since $\Delta_i \cap \Delta_j = \emptyset$ it follows that $(A,\Omega) \neq (A',\Omega')$, and so $V(A \cap A') \subseteq \bar{\Omega} \cap \bar{\Omega}' \subseteq V(\mathcal{S})$; and hence $v \in V(\mathcal{S})$, a contradiction. Thus there is no such v, as required. □

(7.7) *If there is no crooked transaction of cardinality π in (G_i^*,Ω_i^*) for $1 \leq i \leq k$, then (S3) holds.*

Proof. By (6.1), there is a segregation $\mathcal{S}_i$ of G_i^* of type $(3\pi+9,1)$ with $\bar{\Omega}_i^* \subseteq V(\mathcal{S}_i)$, and an arrangement α_i of $\mathcal{S}_i$ in a closed disc Δ, such that $\Omega_i^* = \alpha_i^{-1}(\mathrm{bd}(\Delta))$. By applying a suitable homeomorphism, we may assume that $\Delta = \Delta_i$, and $\alpha_0(v) = \alpha_i(v)$ for each $v \in \bar{\Omega}_i^*$. Let

$$\mathcal{S}' = \{(A,\Omega) \in \mathcal{S}_0 \colon \alpha_0(A,\Omega) \not\subseteq \Delta_1,\ldots,\Delta_k\}.$$

(1) *For $1 \leq i \leq k$, if $(A,\Omega) \in \mathcal{S}'$ then $E(G_i^* \cap A) = \emptyset$ and $V(G_i^* \cap A) \subseteq \bar{\Omega}_i^*$.*

Subproof. Let x be a vertex or edge of G which belongs to $G_i^* \cap A$. Since

$$G_i^* \cap A \subseteq (J_i \cup \alpha_0^{-1}(\Delta_i)) \cap G_0$$

and $J_i \cap G_0 \subseteq \alpha_0^{-1}(\Delta_i)$, it follows that $A \subseteq \alpha_0^{-1}(\Delta_i)$. Suppose that it is not the case that $x \in V(\mathcal{S}_0)$ and $\alpha_0(x) \in \Delta_i$. Then there exists $(A',\Omega') \in \mathcal{S}_i$ with $\alpha_0(A',\Omega') \subseteq \Delta_i$, such that x belongs to A', since x belongs to $\alpha_0^{-1}(\Delta_i)$.

[p. 140] Now $(A',\Omega') \neq (A,\Omega)$ because $\alpha_0(A,\Omega) \not\subseteq \Delta_i$, and so $V(A \cap A') \subseteq \bar{\Omega} \cap \bar{\Omega}'$, since $\mathscr{S}_0$ is a segregation. Hence $x \in \bar{\Omega}' \subseteq V(\mathscr{S}_0)$, and so $\alpha_0(x) \in bd(\alpha_0(A',\Omega')) \subseteq \Delta_i$, contrary to our assumption. Hence $x \in V(\mathscr{S}_0)$ (in particular, x is a vertex), and $\alpha_0(x) \in \Delta_i$. Now since $V(A) \cap V(\mathscr{S}_0) = \bar{\Omega}$, it follows that $x \in \bar{\Omega}$, and so $\alpha_0(x) \in bd(\alpha_0(A,\Omega))$. But $\alpha_0(A,\Omega) \cap \Delta_i \subseteq bd(\Delta_i)$, and $\alpha_0(x) \in \Delta_i$, and so $\alpha_0(x) \in bd(\Delta_i)$. Consequently $x \in \bar{\Omega}_i^*$. This proves (1).

Let $\mathscr{S} = \mathscr{S}_0 \cup \mathscr{S}_1 \cup \cdots \cup \mathscr{S}_k$.

(2) *$\mathscr{S}$ is a segregation of G.*

Subproof. We must show

(i) $A \subseteq G$ for each $(A,\Omega) \in \mathscr{S}$, and $\bigcup(A : (A,\Omega) \in \mathscr{S}) = G$

(ii) for distinct $(A,\Omega),(A',\Omega') \in \mathscr{S}$, $V(A \cap A') \subseteq \bar{\Omega} \cap \bar{\Omega}'$ and $E(A \cap A') = \emptyset$.

We first show (i). Certainly $A \subseteq G$ for each $(A,\Omega) \in \mathscr{S}$. Let x be a vertex or edge of G; to show that $\bigcup(A : (A,\Omega) \in \mathscr{S}) = G$ it suffices to show that x belongs to A for some $(A,\Omega) \in \mathscr{S}$. If x belongs to G_i^* for some $i \geq 1$, then x belongs to A for some $(A,\Omega) \in \mathscr{S}_i \subseteq \mathscr{S}$, as required. If not, then x does not belong to $J_1,\ldots,J_k$, and so x belongs to G_0. Hence there exists $(A,\Omega) \in \mathscr{S}_0$ such that x belongs to A. For $1 \leq i \leq k$, since x does not belong to G_i^* and $\alpha_0^{-1}(\Delta_i) \subseteq G_i^*$, it follows that $\alpha_0(A,\Omega) \not\subseteq \Delta_i$. Hence $(A,\Omega) \in \mathscr{S}$. This proves (i).

For (ii), let $(A,\Omega),(A',\Omega') \in \mathscr{S}$ be distinct. If they both belong to $\mathscr{S}_0$, or for some $i \geq 1$ they both belong to $\mathscr{S}_i$, then (ii) holds since $\mathscr{S}_0,\mathscr{S}_1,\ldots,\mathscr{S}_k$ are segregations. We may assume then that $(A,\Omega) \in \mathscr{S}_i$ for some $i \geq 1$, and $(A',\Omega') \notin \mathscr{S}_i$. If $(A',\Omega') \in \mathscr{S}_j$ for some $j \neq i$, then $A \cap A' \subseteq G_i^* \cap G_j^*$, which is null by (7.6), and (ii) holds. Otherwise, $(A',\Omega') \in \mathscr{S}_0$, and from (1), $E(A \cap A') = \emptyset$ and

$$V(A \cap A') \subseteq \bar{\Omega}_i^* \subseteq V(\mathscr{S}_i) \cap V(\mathscr{S}_0);$$

and since $V(A) \cap V(\mathscr{S}_i) = \bar{\Omega}$ and $V(A') \cap V(\mathscr{S}_0) = \bar{\Omega}'$, it follows that (ii) holds. This proves (2).

(3) *$\mathscr{S}$ is of type (ρ,k).*

Subproof. $\mathscr{S}_0$ is a 3-segregation, and each $\mathscr{S}_i$ is of type $(3\pi+9,1)$, and so $\mathscr{S}$ is of type $(3\pi+9,k)$. This proves (3).

(4) *$\mathscr{S}$ is $\mathscr{T}^*$-central.*

Subproof. Let $(A,\Omega) \in \mathscr{S}$, and let $B \subseteq G$ so that (A,B) is a separation of G and $V(A \cap B) = \bar{\Omega}$. Suppose that there exists $(A^*,B^*) \in \mathscr{T}^*$ with $B^* \subseteq A$. By (2.1) we may assume that (A^*,B^*) has order $\leq 16\pi+19$. Since [p. 141] $\theta > 6\pi+19$ and $H, \mathscr{T}$ is a Σ-span, it follows that $(A^* \cap H, B^* \cap H) \in \mathscr{T}$. Consequently $(A \cap H, B \cap H) \notin \mathscr{T}$, because $(A^* \cap H) \cup (A \cap H) \supseteq (A^* \cap H) \cup (B^* \cap H) = H$, and so $(A,\Omega) \notin \mathscr{S}_0$, because $\mathscr{S}_0$ is $\mathscr{T}$-local. Thus there exists i with $1 \leq i \leq k$ such that $(A,\Omega) \in \mathscr{S}_i$. Then $B^* \subseteq A \subseteq G_i^*$. However, by [5, Theorem (7.3)], there exists $e \in E(B^* \cap H)$ with

$$d(r_i,e) \geq \theta - |V((A^* \cap H) \cap (B^* \cap H))| \geq \theta - |V(A^* \cap B^*)| \geq \theta - 6\pi - 19.$$

Since $e \in E(H)$, there exists a unique $(A',\Omega') \in \mathscr{S}_0$ with $e \in E(A')$; and since $e \in E(G_i^*)$ it follows that $\alpha_0(A',\Omega') \subseteq \Delta_i$. Since α_0 is compatible with H, we deduce that $e \subseteq \Delta_i$, and so $d(r_i,e) \leq 3\beta+6$ by (7.5)(ii). But $3\beta+6 < \theta - 6\pi - 19$, a contradiction. This proves (4).

Define $\alpha(A,\Omega)$ for $(A,\Omega) \in \mathscr{S}$ as follows: if $(A,\Omega) \in \mathscr{S}_0$, let $\alpha(A,\Omega) = \alpha_0(A,\Omega)$; and otherwise, choose i with $1 \le i \le k$ such that $(A,\Omega) \in \mathscr{S}_i$, and let $\alpha(A,\Omega) = \alpha_i(A,\Omega)$. For $v \in V(\mathscr{S})$, define $\alpha(v)$ by: if $v \in \bar{\Omega}$ for some $(A,\Omega) \in \mathscr{S}_0$, let $\alpha(v) = \alpha_0(v)$; and otherwise, choose i with $1 \le i \le k$ such that $v \in \bar{\Omega}$ for some $(A,\Omega) \in \mathscr{S}_i$, and let $\alpha(v) = \alpha_i(v)$.

(5) *For $1 \le i \le k$, if $v \in V(\mathscr{S}_i)$ then $\alpha(v) = \alpha_i(v)$.*

Subproof. Suppose first that $v \in \bar{\Omega}$ for some $(A,\Omega) \in \mathscr{S}_0$. Then $\alpha(v) = \alpha_0(v)$; but by (1), $v \in \bar{\Omega}_i^*$ and so $\alpha_i(v) = \alpha_0(v) = \alpha(v)$ as required. We assume then that there is no such (A,Ω). Choose $1 \le j \le k$ and $(A,\Omega) \in \mathscr{S}_j$ with $v \in \bar{\Omega}$, such that $\alpha(v) = \alpha_j(v)$. Then $v \in V(G_i^* \cap G_j^*)$, and so $i = j$ by (7.6). Hence $\alpha_i(v) = \alpha(v)$. This proves (5).

(6) *$\alpha \in bd(\alpha(A,\Omega))$ for each $(A,\Omega) \in \mathscr{S}$ and each $v \in \bar{\Omega}$.*

Subproof. If $(A,\Omega) \in \mathscr{S}_0$, then $\alpha(A,\Omega) = \alpha_0(A,\Omega)$ and $\alpha(v) = \alpha_0(v) \in bd(\alpha_0(A,\Omega))$, as required. We assume then that $(A,\Omega) \in \mathscr{S}_i$ for some $i \ge 1$. By (5), it follows that $\alpha(v) = \alpha_i(v) \in bd(\alpha_i(A,\Omega))$. This proves (6).

(7) *For distinct $(A,\Omega),(A',\Omega') \in \mathscr{S}$, if $x \in \alpha(A,\Omega) \cap \alpha(A',\Omega')$ then $x = \alpha(v)$ for some $v \in \bar{\Omega} \cap \bar{\Omega}'$.*

Subproof. If $(A,\Omega),(A',\Omega') \in \mathscr{S}_0$, then $x \in \alpha_0(A,\Omega) \cap \alpha_0(A',\Omega')$, and so $x = \alpha_0(v) = \alpha(v)$ for some $v \in \bar{\Omega} \cap \bar{\Omega}'$. We may assume then that $(A,\Omega) \notin \mathscr{S}_0$. Choose i with $1 \le i \le k$ such that $(A,\Omega) \in \mathscr{S}_i$ and $\alpha(A,\Omega) = \alpha_i(A,\Omega) \subseteq \Delta_i$. Suppose that $(A',\Omega') \notin \mathscr{S}_0$. Then there exists j with $1 \le j \le k$ such that $(A',\Omega') \in \mathscr{S}_j$ and $\alpha(A',\Omega') = \alpha_j(A',\Omega')$; and since $\alpha_j(A',\Omega') \subseteq \Delta_j$ and $\alpha_i(A,\Omega) \subseteq \Delta_i$, and so $x \in \Delta_i \cap \Delta_j$, it follows that $i = j$. Hence there exists $v \in \bar{\Omega} \cap \bar{\Omega}'$ with $\alpha_i(v) = x$, and so $\alpha(v) = x$, by (5). We may therefore assume that $(A',\Omega') \in \mathscr{S}_0$. Then $\alpha(A,\Omega) \subseteq \Delta_i$, and $\alpha(A',\Omega') \cap \Delta_i \subseteq bd(\Delta_i)$, and so $x \in bd(\Delta_i)$. Since $bd(\Delta_i)$ is α_0-normal and $x \in \alpha_0(A',\Omega')$, it follows that $x = \alpha_0(v)$ for some $v \in \bar{\Omega}'$; and hence $v \in \bar{\Omega}_i^*$. Consequently $v \in V(\mathscr{S}_i)$, and $\alpha_i(v) = \alpha_0(v) = x$. Since $x \in \alpha_i(A,\Omega)$ we deduce [p. 142] that $v \in \bar{\Omega}$, and so $v \in \bar{\Omega} \cap \bar{\Omega}'$. Moreover, since $v \in \bar{\Omega}'$ it follows that $\alpha(v) = \alpha_0(v)$, and so $\alpha(v) = x$. This proves (7).

(8) *For distinct $v, v' \in V(\mathscr{S})$, $\alpha(v) \ne \alpha(v')$.*

Subproof. Choose $(A,\Omega),(A',\Omega') \in \mathscr{S}$ with $v \in \bar{\Omega}$ and $v' \in \bar{\Omega}'$. If $(A,\Omega),(A',\Omega') \in \mathscr{S}_0$, then $\alpha(v) = \alpha_0(v) \ne \alpha_0(v') = \alpha(v')$ as required. We may therefore assume that $(A,\Omega) \in \mathscr{S}_i$ where $1 \le i \le k$. Consequently, $\alpha_i(v) = \alpha(v)$, by (5). If $v' \in V(\mathscr{S}_i)$ then $\alpha_i(v') \ne \alpha_i(v)$, and $\alpha_i(v') = \alpha(v')$ by (5), and so $\alpha(v) \ne \alpha(v')$ as required. We assume then that $v' \notin V(\mathscr{S}_i)$. If $v' \in V(\mathscr{S}_j)$ for some $j \ne i$ with $1 \le j \le ~~i~~k$,[F18] then by (5), $\alpha(v') = \alpha_j(v') \in \Delta_j$, and so $\alpha(v') \ne \alpha(v)$ because $\alpha(v) \in \Delta_i$. Hence we may assume

[F18] **Misprint.** The next sentence has $1 \le j \le k$.

that $v' \notin V(\mathscr{S}_j)$ for $1 \le j \le k$. Thus $(A', \Omega') \in \mathscr{S}_0$. Since $\bar{\Omega}_i^* \subseteq V(\mathscr{S}_i)$ it follows that $v' \notin \bar{\Omega}_i^*$, and so $\alpha_0(v') \notin bd(\Delta_i)$. But $\alpha_0(v') \in \alpha_0(A', \Omega')$ and $\alpha_0(A', \Omega') \cap \Delta_i \subseteq bd(\Delta_i)$, and so $\alpha_0(v') \notin \Delta_i$. Consequently $\alpha(v') \neq \alpha(v)$, as required. This proves (8).

(9) *For all $(A, \Omega) \in \mathscr{S}$, Ω is mapped by α to a natural order of $\alpha(\bar{\Omega})$ from $bd(\alpha(A, \Omega))$.*

Subproof. This is trivial unless $|\bar{\Omega}| \ge 4$; and if $|\bar{\Omega}| \ge 4$ then $(A, \Omega) \in \mathscr{S}_i$ for some i with $1 \le i \le k$, and $\alpha(A, \Omega) = \alpha_i(A, \Omega)$, and the result follows by (5) since α_i is an arrangement of $\mathscr{S}_i$. This proves (9).

From (6), (7), (8), (9), it follows that α is an arrangement of $\mathscr{S}$ in Σ. From (2), (3) and (4) we deduce that (S3) holds, as required. □

In view of (7.7), we may assume henceforth that (G_i^*, Ω_i^*) has a crooked transaction of cardinality π for some i, say $i = 1$. We recall that Ω_1 is a natural order of $V(H) \cap bd(\Lambda_1)$ from $bd(\Lambda_1)$.

(7.8) *There is a transaction $\mathscr{P}$ in $(G_0 \cup J_1, \Omega_1)$ such that $V(\mathscr{P})$ is free with respect to the $(12\beta+14)$-compression of $\mathscr{T}$ in $H \cap (\Sigma - \Lambda_1)$, each $P \in \mathscr{P}$ meets $H \cap (\Sigma - \Lambda_1)$ precisely in the ends of P, and one of the following holds:*

(i) *$|\mathscr{P}| = \phi$ and $\mathscr{P}$ is a crosscap transaction*

(ii) *$|\mathscr{P}| = \tau_1$ and $\mathscr{P}$ is a leap transaction*

(iii) *$|\mathscr{P}| = \tau_2$ and $\mathscr{P}$ is a doublecross transaction.*

Proof. We are assuming that (G_1^*, Ω_1^*) has a crooked transaction of cardinality π. By (7.5)(v), there is a crooked transaction $\mathscr{P}'$ of cardinality π in $(G_0 \cup J_1, \Omega_1)$ such that $V(\mathscr{P}')$ is free with respect to the $(12\beta + 14)$-compression of $\mathscr{T}$ in $H \cap (\Sigma - \Lambda_1)$, and each $P \in \mathscr{P}'$ meets $H \cap (\Sigma - \Lambda_1)$ precisely in the ends of P. By (6.2), there exists $\mathscr{P} \subseteq \mathscr{P}'$ satisfying (i), (ii) or (iii) above, and hence satisfying the theorem, since any subset of a free set is free by [2, Theorem (12.2)]. □

Henceforth $\mathscr{P}$ is a transaction satisfying (7.8).

[p. 143] (7.9) *If $\mathscr{P}$ is a crosscap transaction of cardinality ϕ then (S2) is true.*

Proof. This is immediate from (3.3). □

Let $\mathscr{T}_1$ be the $(12\beta + 14)$-compression of $\mathscr{T}$ in $H \cap (\Sigma - \Lambda_1)$.

(7.10) *$\mathscr{P}$ is not a leap transaction of cardinality τ_1.*

Proof. Suppose that $\mathscr{P}$ is a leap transaction of cardinality τ_1, and let $\mathscr{P} = \{P_1, \ldots, P_{\tau_1}\}$, where P_i has ends $v_i, v_{2\tau_1 - i}$ for $1 \le i \le \tau_1 - 1$, and P_{τ_1} has ends $v_{\tau_1}, v_{2\tau_1}$, and where $v_1, \ldots, v_{2\tau_1}$ occur in Ω_1 in order. By (3.4) applied to $H \cap (\Sigma - \Lambda_1), \mathscr{T}$, we deduce that there is a Σ-span $H', \eta', \mathscr{T}'$ of order $\theta - 12\beta - 14$, obtained from $H, \mathscr{T}$ by rearranging within β of r_1, and P_{τ_1} is an $\eta'(H')$-path in G, and $d'(v_{\tau_1}, v_{2\tau_1}) \ge 2\tau_1 - 8 = \mu$. Hence $H', \eta', \mathscr{T}'$

is μ-stepped. But $\beta \le \lambda$, and so $H\mathscr{T}$ is (β,μ)-flat by (5.1), a contradiction. Thus there is no such $\mathscr{P}$. □

(7.11) *If $\mathscr{P}$ is a doublecross transaction of cardinality τ_2 then (S1) holds.*

Proof. For convenience, we write τ for τ_2. Let $\mathscr{P} = \{P_1, \dots, P_\tau\}$, where P_i has ends $v_i, v_{2\tau-3-i}$ for $1 \le i \le \tau-4$, and $P_{\tau-3}$ has ends $v_{2\tau-3}, v_{2\tau-1}$, $P_{\tau-2}$ has ends $v_{2\tau-2}, v_{2\tau}$, $P_{\tau-1}$ has ends $v_{\tau-3}, v_{\tau-1}$ and P_τ has ends $v_{\tau-2}, v_\tau$. By "drawing" $P_1, \dots, P_{\tau-4}$ in $\bar{\Lambda}_1$, we see that there is a drawing H' in Σ with $U(H') \cap bd(\Lambda_1) \subseteq V(H')$, such that $H' \cap (\Sigma - \Lambda_1) = H \cap (\Sigma - \Lambda_1)$, and there is an isomorphism η' from H' to a subgraph of G, such that $\eta'(x) = x$ for every vertex or edge in $H'(\Sigma - \Lambda_1)$, and

$$\eta'(H') = (H \cap (\Sigma - \Lambda_1)) \cup P_1 \cup \cdots \cup P_{\tau-4}.$$

Let $Q_1, \dots, Q_{\tau-4}$ be the paths of H' such that $\eta'(Q_i) = P_i (1 \le i \le \tau-4)$. Let s_1 be the region of H' in Λ incident with $v_{2\tau-4}, v_{2\tau-3}, \dots, v_{2\tau}, v_1$; and let s_2 be the region incident with $v_{\tau-4}, v_{\tau-3}, \dots, v_{\tau+1}$. Let $\mathscr{T}'$ be the tangle in H' induced by $\mathscr{T}_1$. By (3.4) applied to $H \cap (\Sigma - \Lambda_1), H'$, we see that

(1) *$H', \eta', \mathscr{T}'$ is a Σ-span of order $\theta - 12\beta - 14$, and $d'(s_1, s_2) \ge 2\tau - \not{6}12 = \theta'$.*[F19]

(2) *s_1 and s_2 are eyes of $H', \eta', \mathscr{T}'$.*

Subproof. Now $\{v_1, \dots, v_{2\tau}\}$ is free with respect to $\mathscr{T}_1$, and hence with respect to $\mathscr{T}'$. Since any subset of a free set is free, it follows that $\{v_{2\tau-3}, v_{2\tau-2}, v_{2\tau-1}, v_{2\tau}\}$ is free with respect to $\mathscr{T}'$. The region s_1 of H' is bounded by a circuit formed by P_1 and part of the circuit bounding Λ_1; and hence, because of the $\eta'(H')$-paths $P_{\tau-3}, P_{\tau-2}$, it follows that s_1 is an eye. Similarly s_2 is an eye. This proves (2).

(3) *$r_2, \dots, r_k$ are eyes of $H \cap (\Sigma - \Lambda_1), \mathscr{T}_1$.*

[p. 144] *Subproof.* Certainly $r_2, \dots, r_k$ are regions of $H \cap (\Sigma - \Lambda_1)$, since Λ_1 is a $(3\beta+3)$-zone (with respect to $\mathcal{T}$) around r_1, and $d(r_1, r_i) = \theta > 3\beta + 3$ for $i \ge 2$. Let $i \ge 2$, and let $a, b, c, d \in V(H) \cap \mathrm{bd}(r_i)$ such that $\{a, b, c, d\}$ is free with respect to $\mathcal{T}$. We claim it is free with respect to $\mathcal{T}_1$. For $\mathcal{T}_1$ is a $(4(3\beta+3)+2)$-compression of $\mathcal{T}$, and

$$d(r_1, v) \ge d(r_1, r_i) - 1 = \theta - 1 > 8 + 5(3\beta + 3) + 2$$

for all $v \in \{a, b, c, d\}$; and $x \in A(H \cap (\Sigma - \Lambda_1))$ for every $x \in A(H)$ with $d(r_1, x) > 3\beta + 3$, since Λ_1 is a $(3\beta+3)$-zone. Consequently, by (4.7), it follows that $\{a, b, c, d\}$ is free with respect to $\mathcal{T}_1$. This proves (3).

[F19] **Proof gap.** The value changes together with τ_2 on page 138 (see the note there). Here (3.4) is applied with $k = \tau - 3$, so it gives $2\tau - 12$.

(4) *$r_2, \ldots, r_k$ are eyes of $H', \eta', \mathcal{T}'$.*

Subproof. Let $2 \le i \le k$. Since r_i is an eye of $H \cap (\Sigma - \Lambda_1), \mathcal{T}_1$, there are two disjoint $H \cap (\Sigma - \Lambda_1)$-paths P, Q with ends in $\mathrm{bd}(r_i)$ in alternating order. We claim that P, Q are $\eta'(H')$-paths. For if P is not an $\eta'(H')$-path, then P meets one of $P_1, \ldots, P_{\tau-4}$, and so there is an $H \cap (\Sigma - \Lambda_1)$-path in G with one end, u, in $\mathrm{bd}(r_i)$ and the other, v, in $\mathrm{bd}(\Lambda_1)$. Since

$$d_1(u, v) \ge d_1(r_i, v) - 1 \ge d(r_i, v) - (12\beta + 14) - 1$$

and

$$d(r_i, v) \ge d(r_i, r_1) - d(r_1, v) = \theta - d(r_1, v) \ge \theta - (3\beta + 3)$$

it follows that $d_1(u, v) \ge \theta - 15\beta - 18 \ge \mu$, and so $H \cap (\Sigma - \Lambda_1), \mathcal{T}_1$ is μ-stepped. But $H \cap (\Sigma - \Lambda_1), \mathcal{T}_1$ is obtained from $H, \mathcal{T}$ by rearranging within $3\beta + 3$ of r_1, and $3\beta + 3 \le \lambda$, and $H, \mathcal{T}$ is (λ, μ)-flat, a contradiction. This proves that P and similarly Q are $\eta'(H')$-paths, and hence proves (4).

(5) *For $2 \le i < j \le k$, $d'(r_i, r_j) \ge \theta'$.*

Subproof. $d'(r_i, r_j) \ge d(r_i, r_j) - 12\beta - 14 = \theta - 12\beta - 14 \ge \theta'$.

(6) *For $2 \le i \le k$, $d'(s_1, r_i) \ge \theta'$ and $d'(s_2, r_i) \ge \theta'$.*

Subproof. $d'(s_1, r_i) \ge d_1(\Lambda_1, r_i) \ge d_1(r_1, r_i) - 12\beta - 14 = \theta - 12\beta - 14 \ge \theta'$. Similarly $d'(s_2, r_i) \ge \theta'$.

(7) *$H', \eta', \mathcal{T}'$ is (λ', μ)-flat.*

Subproof. Now $3\beta + 3, \lambda' \ge 4$ and $\mu \ge 1$, and $\lambda \ge 6(3\beta + 3) + 5\lambda' + 3\mu + 4$. Since $H', \eta', \mathcal{T}'$ is obtained from $H, \mathcal{T}$ by rearranging within $3\beta + 3$ of r_1, and $H, \mathcal{T}$ is (λ, μ)-flat, the result follows from (5.2). This proves (7).

Let $\mathcal{T}''$ be the θ'-truncation of $\mathcal{T}'$. Then $H', \eta', \mathcal{T}''$ is a Σ-span of order θ'. Since $\theta' \ge 4$ it follows from (2) and (4) that $s_1, s_2, r_2, \ldots, r_k$ are eyes [p. 145] of $H', \eta', \mathcal{T}''$. From (1), (5), and (6) it follows that for distinct $r, r' \in \{s_1, s_2, r_2, \ldots, r_k\}$,

$$d''(r, r') = \min(\theta', d'(r, r')) = \theta'.$$

By (7) and (5.1), $H', \eta', \mathcal{T}''$ is (λ', μ)-flat, since $\theta' \ge 4\lambda' + \mu + 2$. Hence (S1) holds, as required. □

In view of (7.8), (7.9), (7.10) and (7.11) it follows that in all cases one of (S1), (S2), (S3) holds, and the proof of (1.2) is complete. Hence so is the proof of (1.1).

8. Some Mild Improvements

In order to facilitate applying our main theorem (1.1), we make in this section two modifications to it. The first is concerned with replacing (1.1)(i)

with a condition involving large clique minors, because our application is to graphs with no large clique minor. Thus, let $\mathcal{T}^*$ be a tangle in a graph G, and let $p \geq 0$ be an integer. We say that $\mathcal{T}^*$ controls a K_p minor of G if $\mathcal{T}^*$ has order $\geq p$ and there are p non-null disjoint connected subgraphs $X_1, \ldots, X_p$ of G such that

(i) for $1 \leq i < j \leq p$ there is an edge of G with one end in $V(X_i)$ and the other in $V(X_j)$, and

(ii) there is no $(A, B) \in \mathcal{T}^*$ of order $< p$ such that $V(X_i) \subseteq V(A)$ for some i $(1 \leq i \leq p)$.

[4, Theorem (4.5)] implies (since the tangle in G induced by $\mathcal{T}$ is a subset of $\mathcal{T}^*$) that

(8.1) *For any surface Σ with $bd(\Sigma) = \emptyset$, and any integer $p \geq 0$ there exists $\kappa, \phi \geq 0$ such that the following is true. Let $\mathcal{T}^*$ be a tangle in a graph G of order $\geq p$, and let $H, \eta, \mathcal{T}$ be a Σ-span of order $\geq \phi$, with $\geq \kappa$ independent eyes. Then $\mathcal{T}^*$ controls a K_p minor of G.*

Hence we may modify (1.1) as follows.

(8.2) *For any surface Σ with $bd(\Sigma) = \emptyset$, and any integers $p, \phi, \mu \geq 0$, there are integers $\kappa, \theta, \lambda, \rho \geq 0$ such that the following holds. Let $\mathcal{T}^*$ be a tangle in a graph G such that there is no $\mathcal{T}^*$-central segregation of G of type (ρ, κ) with an arrangement in Σ. Then either*

[p. 146] (i) *$\mathcal{T}^*$ controls a K_p minor of G, or*

(ii) *there is a Σ'-span of order $\geq \phi$, where Σ' is a surface obtained by adding a crosscap to Σ, or*

(iii) *no Σ-span of order $\geq \theta$ is (λ, μ)-flat.*

Proof. If the result holds for $4, \phi, \mu$ then it holds for p, ϕ, μ for all $p \leq 4$, because if $\mathcal{T}^*$ controls a K_4 minor then it controls a K_p minor for all $p \leq 4$. Hence we may assume that $p \geq 4$. Let κ, ϕ_1 satisfy (8.1) (with ϕ replaced by ϕ_1), and let $\phi_2 = \max(\phi, \phi_1)$. Let θ, λ, ρ satisfy (1.1) (with ϕ replaced by ϕ_2). We may assume that $\theta \geq p$. Then we claim $\kappa, \theta, \lambda, \rho$ satisfy (8.2). For let $\mathcal{T}^*, G$ be as in the theorem. If $\mathcal{T}^*$ has order $< p$ then (8.2)(iii) holds, since $\theta \geq p$. Thus we may assume that $\mathcal{T}^*$ has order $\geq p$. By (1.1), one of (1.1)(i), (ii), (iii) holds. If (1.1)(i) holds then (8.2)(i) holds, by (8.1), and if (1.1)(ii) or (iii) holds, then (8.2)(ii) or (iii) holds. □

One of the conclusions of (8.2) is that no Σ-span of order $\geq \theta$ is (λ, μ)-flat, and the second modification we wish to make concerns this. If $\mathcal{T}^*$ is a tangle in G and $H, \eta, \mathcal{T}$ is a Σ-span of order θ, where Σ is a surface with $\mathrm{bd}(\Sigma) = \emptyset$, we say that $H, \eta, \mathcal{T}$ is (λ, μ)-level if

(i) $\theta \geq 4\lambda + \mu + 2$

(ii) $H, \eta, \mathcal{T}$ is not μ-stepped, and

(iii) for every Σ-span $H', \eta', \mathcal{T}'$, obtained from $H, \eta, \mathcal{T}$ by rearranging within λ of some $z \in A(H)$, there is no $\eta'(H')$-path in G with ends $\eta'(u), \eta'(v)$ with $d'(u,v) \geq \mu$ and $d(z,u), d(z,v) \leq \lambda$.

(The difference between (i)–(iii) here and the definition of (λ, μ)-flat is the requirement that $d(z,u), d(z,v) \leq \lambda$.)

(8.3) *Let $\lambda, \mu \geq 0$ be integers, let $\mathcal{T}^*$ be a tangle in a graph G, and let Σ be a surface with $bd(\Sigma) = \emptyset$. Then every Σ-span which is $(\lambda+\mu+2, \mu)$-level is (λ, μ)-flat.*

Proof. Suppose that $H, \eta, \mathcal{T}$ is a Σ-span of order θ is not (λ, μ)-flat; we shall show that it is not $(\lambda + \mu + 2, \mu)$-level. If $\theta < 4(\lambda + \mu + 2) + \mu + 2$ then $H, \eta, \mathcal{T}$ is not $(\lambda + \mu + 2, \mu)$-level as required, and so we assume that $\theta \geq 4\lambda + 5\mu + 10$. Since $H, \eta, \mathcal{T}$ is not (λ, μ)-flat and $\theta \geq 4\lambda + \mu + 2$, there is a μ-stepped Σ-span $H', \eta', \mathcal{T}'$ of order $\theta - 4\lambda - 2$, obtained from $H, \eta, \mathcal{T}$ by rearranging within λ of some $z \in A(H)$. Let P be an $\eta'(H')$-path with ends $\eta'(a), \eta'(b)$, such that $d'(a,b) \geq \mu$.

[p. 147] (1) *If $d(z,a), d(z,b) \leq \lambda+\mu+2$, then $H, \eta, \mathcal{T}$ is not $(\lambda+\mu+2)$-level.*

Subproof. Let $\mathcal{T}''$ be the $(\theta - 4(\lambda + \mu + 2) - 2)$-truncation of $\mathcal{T}'$; then $H', \eta', \mathcal{T}''$ is a Σ-span of order $\theta - 4(\lambda + \mu + 2) - 2$, obtained from $H, \eta, \mathcal{T}$ by rearranging within $\lambda + \mu + 2$ of z, and P is an $\eta'(H')$-path, and

$$d''(a,b) = \min(\theta - 4(\lambda + \mu + 2) - 2, d'(a,b)) \geq \mu,$$

and so condition (iii) in the definition of "level" is false. This proves (1).

From (1) we may, therefore, assume that

(2) *$d(z,a) \geq \lambda + \mu + 3$, and $a \in V(H)$, and $\eta(a) = \eta'(a)$.*

Let e be the edge of P incident with $\eta(a)$.

(3) *e is not an edge of $\eta(H)$.*

Subproof. Suppose that $e = \eta(f)$ for some $f \in E(H)$. Since $\eta(f)$ is incident with $\eta(a)$ it follows that a is an end of f in H, and so $d(a,f) \leq 2$. From (2), it follows that $d(z,f) \geq \lambda + \mu + 1$, and so $f \in E(H')$ and $\eta'(f) = \eta(f) = e$. But $e \in E(P)$, and P is an $\eta'(H')$-path, a contradiction. This proves (3).

Let Λ be the union of all $x \in A(H)$ with $d(z,x) \leq \lambda$. We may assume that

(4) *There is no $v \in V(H)$ with $v \neq a$ such that $\eta(v) \in V(P)$.*

Subproof. Suppose that there is such a vertex, and choose it so that the subpath Q of P between $\eta(a)$ and $\eta(v)$ is minimal. Then Q is an $\eta(H)$-path, by (3). If $d(a,v) \geq \mu$ then $H, \eta, \mathcal{T}$ is μ-stepped and so not $(\lambda+\mu+2, \mu)$-level, as required, and so we suppose, for a contradiction, that $d(a,v) \leq \mu - 1$.

Consequently, $d(z,v) \geq \lambda + 4$ and so $v \in V(H')$ and $\eta'(v) = \eta(v)$. Since P is an $\eta'(H')$-path it follows that $Q = P$, $v = b$, and $d(a,b) \leq \mu - 1$. Since $d(z,a) \geq \lambda + \mu + 3$, and $d(z,\sigma) \leq \lambda$ for all $\sigma \in \Lambda$, it follows that $d(a,\sigma) \geq \mu + 3 > d(a,b)$ for all $\sigma \in \Lambda$. By (4.2), we deduce that $d'(a,b) \leq d(a,b) < \mu$, a contradiction. We may, therefore, assume that (4) holds.

From (4), it follows that $b \in \Lambda$; for otherwise $b \in V(H)$ and $\eta(b) = \eta'(b) \in V(P)$, contrary to (4). Since H' is 2-cell and hence connected, there is a path of H' from b to a, and hence we may choose a minimal path Q of $\eta'(H')$ from $\eta'(b)$ to $V(\eta(H))$. Let Q have ends $\eta'(b), \eta(c)$ where $c \in V(H)$. It follows that no edge of Q is in $E(\eta(H))$, and no vertex of Q is in $V(\eta(H))$ except $\eta(c)$.

(5) *Q has at least one edge.*

Subproof. Otherwise $\eta'(b) = \eta(c)$, and so $\eta(c) \in V(P)$. By (4), $c = a$, and so $\eta'(b) = \eta(c) = \eta(a) = \eta'(a) \neq \eta'(b)$, a contradiction. This proves (5).

[p. 148] (6) $d(z,c) \leq \lambda + 1$.

Subproof. Let $f \in E(H')$ be such that $\eta'(f)$ is the edge of Q incident with $\eta(c)$. If $f \not\subseteq \Lambda$, then $f \in E(H)$ and $\eta(f) = \eta'(f) \in E(Q)$, a contradiction since no edge of Q is in $E(\eta(H))$. Thus $f \subseteq \Lambda$, and so $c \in \bar{\Lambda}$. Since $c \in V(H)$ there exists $\sigma \in \Lambda$ with $d(c,\sigma) \leq 1$, and consequently $d(z,c) \leq d(z,\sigma) + 1 \leq \lambda + 1$. This proves (6).

From (6) we deduce that $c \neq a$, since $d(z,a) > \lambda' \geq \lambda + 1$, and so $P \cup Q$ is an $\eta(H)$-path with ends $\eta(a), \eta(c)$. From (2) and (6),

$$d(a,c) \geq d(z,a) - d(z,c) \geq (\lambda + \mu + 3) - (\lambda + 1) = \mu + 2 \geq \mu,$$

and so $H, \eta, \mathscr{T}$ is μ-stepped and not $(\lambda + \mu + 2, \mu)$-level. This completes the proof. □

From (8.2) and (8.3) we obtain

(8.4) *For any surface Σ with $bd(\Sigma) = \emptyset$, and any integers $p, \phi, \mu \geq 0$, there are integers $\kappa, \theta, \lambda, \rho \geq 0$ such that the following holds. Let $\mathscr{T}^*$ be a tangle in a graph G such that there is no $\mathscr{T}^*$-central segregation of G of type (ρ, κ) with an arrangement in Σ. Then either*

(i) *$\mathscr{T}^*$ controls a K_p minor of G, or*

(ii) *there is a Σ'-span of order $\geq \phi$, where Σ' is a surface obtained by adding a crosscap to Σ, or*

(iii) *no Σ-span of order $\geq \theta$ is (λ, μ)-level.*

Proof. Choose $\kappa, \theta, \lambda', \rho$ to satisfy (8.2). Let $\lambda = \lambda' + \mu + 2$; and we claim that (8.4) holds. For let $\mathscr{T}^*, G$ be as (8.4). By (8.2), we may assume that

no Σ-span of order $\geq \theta$ is (λ', μ)-flat. By (8.3), no Σ-span of order $\geq \theta$ is (λ, μ)-level, as required. $\square$

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